Preface

This is the summary of three years of work spanning multiple areas. From the problem to the solution. It is inevitable to name some tools and concepts before they are explained later on among chapters. The project described herein is under active development and the contents of this document may not be completely up-to-date or they may already include fixes and update that are going to be deployed in the short term.

Notation

Given a set $\mathscr{U}$, $\mathrm{int}(\mathscr{U})$ denotes its interior. The set $\mathbb{N}$ is defined as containing all the positive integer numbers, excluding 0, whereas $\mathbb{Z}$ denotes the set of all positive and negative integer numbers, including 0. The sets of real and complex numbers are denoted $\mathbb{R}$ and $\mathbb{C}$, respectively.

The imaginary unit is denoted as j. Given a complex number $z \in \mathbb{C}$, $\mathrm{Re}(z)$ and $\mathrm{Im}(z)$ denote its real and imaginary parts, respectively. Define the following sets:

- $\mathbb{Z}_{\geq 0} = \{k \in \mathbb{Z} : k \geq 0\}$;
- $\mathbb{R}_{\geq 0} = \{a \in \mathbb{R} : a \geq 0\}$;
- $\mathbb{R}_{>0} = \{a \in \mathbb{R} : a > 0\}$;
- $\mathbb{C}_0 = \{z \in \mathbb{C} : \mathrm{Re}(z) = 0\}$;

- $\mathbb{C}_{<0} = \{z \in \mathbb{C} : \mathrm{Re}(z) < 0\}$;
- $\mathbb{C}_{\geq 0} = \{z \in \mathbb{C} : \mathrm{Re}(z) \geq 0\}$.

Given a vector $v \in \mathbb{R}^n$, the symbol $\|v\|_2$ denotes its Euclidean norm. Given a vector space V, $\dim(V)$ denotes its dimension. e_j denotes the j-th vector of the canonical basis of $\mathbb{R}^n$. The expression $\|v\|_R^2$ denotes the product $v'Rv$.

Given a matrix A, $\mathrm{Im}(A)$ denotes its image, $\ker(A)$ denotes its null space, A' denotes its transpose, $A^\dagger$ denotes its Moore-Penrose inverse, and $A^\perp$ denotes a basis matrix for its null space. $\mathrm{spec}(A)$ denotes the spectrum of the matrix A, and $\mathrm{rank}(A)$ denotes its rank. $\vec{(}A)$ denotes the vectorization of the matrix A. Given two matrices $A, B \in \mathbb{R}^{m\times n}$ $A \odot B$ denotes their Hadamard, or entrywise, product, while $A \otimes B$ denotes their Kronecker product. Given the square matrices $\{A_1, \ldots, A_n\}$, the operator $\mathrm{diag}(A_1, \ldots, A_n)$ denotes the square matrix obtained by concatenating the former matrices on its diagonal, while the operator $\mathrm{row}(A_1, \ldots, A_n)$ denotes the matrix obtained by concatenating the former matrices on a row.

The symbol I_n denotes the identity matrix of order n, and 0_n is a matrix of zeroes of order n; the subscripts are omitted when clear from the context. For a block-partitioned matrix M, $M_{[k,h]}$ is its $(k,h)-$th block, $M_{[:,h]}$ is its $h-$th block column, and $M_{[:,1:h]}$ is the sub-matrix formed by its first h block columns.

Given a function $f(x)$, $f : \mathbb{R}^n \to \mathbb{R}$, ∇f denotes its gradient as a column vector, whereas $\frac{\partial f}{\partial x}$ denotes its jacobian. $\mathscr{L}_2([a,b])$, $a, b \in \mathbb{R}$, denotes the set of all the functions that are square-integrable over the interval $[a,b]$.

For $T \in \mathbb{R}_{>0}$ and $t \in \mathbb{R}$, the notation $\mathrm{mod}(t,T)$ denotes the value of t modulo T.

Let $k \in \mathbb{Z}_{\geq 0}$ be the discrete time variable. Let $t \in \mathbb{R}_{\geq 0}$ be the continuous time variable. Given a signal $v(t)$, v_k denotes its value at time $t = k\tau_s$, *i.e.*, $v_k = v(k\tau_s)$, where $\tau_s \in \mathbb{R}_{>0}$ is the sampling time. Consider the continuous-time dynamic system Σ defined by the equation

$$\dot{x}(t) = f(x(t)), \quad t \in \mathbb{R}_{\geq 0},$$

where $x(t) \in \mathbb{R}^n$ is the state vector and the function $f : \mathbb{R}^n \to \mathbb{R}^n$ is smooth. Let $x(0) = x_0$, $x_0 \in \mathscr{U}$. Then, for any $x_0 \in \mathscr{U}$, the solution of the system Σ is uniquely defined for $t \in [0, T)$, where $T = T(x_0)$, and it can be expressed by a function $x(t) = \Phi_t^f(x_0)$. The function $\Phi_t^f : \mathscr{U} \to \mathbb{R}^n$ is termed *flow* of f. Moreover, the system Σ is said to be *complete* if and only if its solution exists for any $t \in \mathbb{R}_{\geq 0}$.

For $s \in \mathbb{C}$, let $W(s)$ be the transfer function of a continuous-time, linear time-invariant system; $\mathscr{P}(W(s))$ denotes the set of poles of $W(s)$.

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7 Plasma shape control for tokamaks

In Chapter 6, the second application scenario of this work has been introduced. In particular, Section 6.1 illustrated the Tokamak à Configuration Variable device, and Section 6.2 provided a thorough description of the Système de Contrôle Distribué that controls TCV. That description clarified how the distributed control system of TCV allows multiple diagnostics, algorithms, estimators, and controller to coexist and interact within a single control architecture. Thanks to this design, the experimental capabilities of TCV have been extended significantly. One of the latest innovations in this context has been the introduction of a *shape controller*, *i.e.*, a subsystem that acts as a reference governor and PID controller for the coil power supplies in order to track a desired plasma shape. Its implementation currently works in coordination with other SCD subsystem, in particular those that generate feedback measurements concerning the equilibrium reconstruction; the measurement that they provide are used to evaluate the shape control law in real-time.

Plasma shapes, characterized by distinct cross-sectional geometries in the poloidal plane, exhibit varying properties regarding stability, performance, and interaction with vessel components. The plasma shape significantly influences the stability characteristics of global *magnetohydrodynamic* (MHD) modes and regulates particle transport phenomena. Furthermore, precise plasma shaping enables an optimal use of both the chamber and the plasma volumes while facilitating controlled management of power deposition from plasma-wall interactions in regions where energy escapes the confined plasma core.

The redundant array of poloidal field coils makes TCV a unique device in the tokamak landscape. The flexibility of the PF coil system allows the researchers to explore a wide range of plasma shapes, each of which exhibits different phenomena and is relevant for the advancement of both plasma and fusion research. This capability has been exploited extensively during the thirty years of TCV operation, leading to significant results in the field [1, 2, 3, 4].

However, regarding the SCD implementation, a real shape control subsystem of the SCD magnetic control has never been fully developed and made operational until 2024. Before, the shape control function has been achieved in feedforward mode, *i.e.*, by evaluating offline the coil current references to give to the power supplies during each shot. This procedure is certainly prone to errors, as it has no feedback and does not take into account real-time measurements of the plasma shape. However, it worked in most cases thanks to the high quality of the offline equilibrium reconstruction and the very brief duration of TCV shots. Nonetheless, to fully exploit the experimental capabilities of the device and to ensure an effective and reliable control of the plasma shape, a real-time shape controller is necessary.

This chapter presents the work that has been done on the implementation of the TCV shape controller while working on the SCD. First, the current state of the art in shape control of tokamak plants is reviewed, focusing subsequently on what has been done recently on TCV. Then, the limitations of the current implementation are discussed, and a set of possible improvements is evaluated. Finally, an improved scheme is designed and evaluated in dedicated experiments.

7.1 State of the art in plasma shape control

The plasma-circuit interaction dynamics are fundamentally governed by nonlinear partial differential equations (PDEs). However, control system design typically employs reduced-order, linear time-invariant models based on ordinary differential equations (ODEs). This methodological gap is addressed through specialized plasma modeling software suites that generate simplified mathematical representations while preserving essential physical characteristics. These computational tools, termed *codes*, enable a systematic approach to model reduction, striking a balance between retaining the fundamental plasma physics and producing tractable formulations suitable for control synthesis.

To define a plasma shape and its position, two key elements are considered: the *centroid* and the *last closed flux surface* (LCFS). The centroid is the center of the plasma cross-section along the poloidal plane, and can be either *magnetic* or *geometric*; the type of centroid to consider depends on the specific application at hand. The magnetic centroid is usually involved when it is convenient to approximate the plasma as a filament carrying the plasma current; in that case, the magnetic centroid corresponds to this ideal filament revolving around the vessel. When dealing with magnetic control, which is also the case of this work, the magnetic centroid is usually considered.

Poloidal field coils, running along the torus, generate a poloidal field that induces a *poloidal flux* in the vessel. Along the plasma volume, there are points in which this flux has the same value. Examining the plasma from a poloidal cross-section, one can group this points by flux values, thus identifying *isoflux surfaces*. These run outward starting from the plasma centroid, initially as closed surfaces that enclose one another. The last closed one of these surfaces is identified as the LCFS.

Thus, controlling the plasma shape usually involves tracking reference setpoints for the centroid and the LCFS, expressed as a set of control points. The shape control capabilities directly impact operational parameters and plasma-material interactions, allowing for advanced scenarios that maximize performance while preserving machine integrity. This type of control represents a critical tool for achieving optimal plasma conditions while maintaining operational safety margins. To achieve different plasma configurations in terms of shape, the machine must be equipped with both a vessel large enough to contain them and with a set of coils that can generate the necessary magnetic fields.

Plasma shape control techniques can be broadly classified into two families: *isoflux-based control* and *gap control*. In isoflux-based control techniques [5], a set of shape control points identifies the last closed flux surface, and the poloidal flux and magnetic field at these points are regulated to track the desired plasma shape. This approach is direct and yields a high tracking precision, requiring minimal modification to the equilibrium reconstruction code as it relies solely on the interpolation of flux and field maps at the control point locations. By minimizing discrepancies between measured and desired flux profiles, isoflux control ensures that the plasma maintains optimal stability and performance.

Conversely, gap control [6] focuses on maintaining fixed distances between the plasma surface boundary and the surrounding vacuum vessel or other wall components. These distances are maintained by acting on the magnetic fields generated by the coils. The control laws involved operate in feedback from the measurements of magnetic probes that evaluate the gaps at some *critical* points. Gap control proves particularly effective in ensuring that the plasma does not approach the vessel wall too closely, thereby reducing the risk of increased impurity influx, excessive cooling of the plasma edge, or even direct physical contact with the vessel.

Regulating plasma shape can be conceptualized as a Multi-Input Multi-Output (MIMO) control problem, which is commonly addressed using model-based methodologies that rely on linearized plasma response models, such as the CREATE-L [7] model. A renown application example of this approach is the eXtreme Shape Controller (XSC), originally developed and validated for the JET tokamak [8]. Over time, controllers with a similar structural design have been implemented in various other devices, including the EAST [5] and the JT-60SA [9] tokamaks.

7.2 State of the art in plasma shape control at TCV

Despite TCV's redundant coil configuration and the broad range of plasma shapes it can support, the absence of a unified shape control system that fully integrates with existing shot design and preparation tools has remained a persistent limitation. Previous attempts at developing a shape controller for TCV have shed light on both the complexity of the task and the potential rewards brought by a successful solution. Early work reported in [10, 11] introduced an H_∞ controller aimed at managing key global shape parameters: triangularity, elongation, and vertical displacement. Although this approach represented a significant step forward, it relied heavily on a control system that is no longer in use at TCV, preventing its applicability in contemporary experiments. Another notable effort described in [12] proposed a novel shape control design but entirely replaced the pre-existing magnetic control architecture. While conceptually appealing, this approach hindered seamless integration with the established framework and shot planning methodologies (see Section 6.2), ultimately reducing its practical value.

More recently, a pioneering AI-based plasma shape controller developed in collaboration with Google DeepMind [13] demonstrated remarkable capabilities during discharges. It has been developed to act on each phase of a shot, from the breakdown to the shape formation; it is currently being expanded to achieve more sophisticated plasma configurations, like doublets. This innovative system highlighted the promise of advanced computational methods for achieving more agile and responsive control. However, its dependence on extensive computational resources (particularly for neural network tuning) has made it challenging to adapt and reconfigure between experiments.

The current approach is described in [14]. It implements an isoflux control strategy to regulate the plasma boundary configuration. This approach maintains the plasma shape by enforcing equivalent poloidal flux values between a set of control points positioned along the last closed flux surface and a reference flux value measured at a specified *boundary point*. The designation of this boundary point differs based on the plasma configuration: for *limiter* plasmas, it corresponds to the limiter contact point, while for *diverted* configurations, it is defined by the primary *X-point* location. The control methodology varies according to the configuration type. In limiter scenarios, the controller enforces a zero orthogonal field component at the contact point to ensure the boundary intersects with the reference position. Diverted configurations, conversely, require simultaneous regulation of both radial and vertical field components to zero at the X-point location, thereby maintaining the desired magnetic topology. Furthermore, the study of some specific divertor configurations requires the control of other quantities such as poloidal flux at non-boundary points, flux difference between active and inactive X-points, and the position of inactive X-points; these too can be controlled by the shape controller.

Once a desired shape has been selected, the first step of the shape controller design procedure consists in determining a linearized model of the TCV plant around the prescribed equilibrium. This is quite a crucial and challenging task, usually solved by the suite of equilibrium reconstruction codes developed at SPC. The result of this design stage also yields a numerical representation of the plasma shape in terms of its LCFS, encoded as a set of control points.

The controller architecture implements a decoupling strategy for the controlled channels at low frequencies through singular value decomposition (SVD) of the steady-state plant dynamics. This decomposition is performed on the weighted static gain matrix of the system. To ensure both numerical feasibility and prevention of excessive control action, the controller design considers only singular values exceeding a specified threshold. This reduction in the problem dimensionality serves the dual purpose of maintaining computational feasibility while limiting control effort.

The architecture incorporates a first-order low-pass filter for each channel to attenuate oscillatory behavior. This design choice addresses the resonance phenomena observed in the Bode plots of the dominant singular modes across various TCV operational scenarios. Control of the selected dominant modes is achieved through a diagonal multiple-input multiple-output (MIMO) PID controller. When a state-space representation of the hybrid controller is available, two approaches exist for gain tuning: an analytical method based on assumed ideal channel decoupling with specified cutoff frequency and phase margin requirements per channel, or optimization-based techniques for refined parameter selection. The PID tuning methodology enables complete automation of the controller design process, upon the availability of a linearized plasma response model. The end user only has to specify the weighting matrices along with fundamental control parameters, such as the desired crossover frequency and phase margin.

This degree of automation proves particularly valuable during experiments, allowing the operators to be effective without extensive control theory expertise and ensuring an easy integration within the existing SCD framework.

7.3 Limitations and possible improvements of the TCV shape controller

The current design of the TCV shape controller has been quite the challenge for the development team, and its integration has turned out to be a delicate matter. Nonetheless, the result is a system that works: given a reference plasma shape and after some tuning, the TCV shape controller is able to control the plasma shape within the limits of the actuators and the diagnostics. Recalling the discussion of the redundant coil configuration of TCV in Section 6.1, which constitutes probably its most peculiar feature, the question arises whether the current shape controller exploits this redundancy to improve the performance of the coil current control or not. Up to this point, it does not.

The PID formulation of the shape controller, thoroughly described in [14], does not account for the actuator redundancy offered by the TCV model, which could in turn be exploited to improve the performance of the shape controller itself. Performance indicators to be considered in this context are the overall power consumption of the coils, as well as the saturation limits of the power supplies. In practical experiments, these have turned out to be important aspects to consider when working with TCV.

The above mentioned question leads one to think whether something can be done to improve the present shape control scheme. This is the starting point for the novel work presented in this chapter. An immediate solution is to include the performance optimization goals in the design of the shape controller. This approach, although ideally feasible, leads to the definition of a control problem that is much different from the original one. Determining a solution to it, considering also all the intricacies and complications that are involved in the control of a plasma shape on TCV, is a challenging task. Another approach consists in keeping the current shape controller design, which works and has been validated through numerous experiments, essentially as it is. However, a new control problem can be formulated at this point, considering the current shape controller as part of the dynamic system and with the goal of optimizing the performance of the shape controller by altering its output before it is fed to the plant. This is the approach that has been followed in this work, relying on a specific subject of control theory that is receiving increasing interest in recent years: that of *dynamic control allocation*.

Dynamic control allocation techniques are used to distribute the control effort among a set of redundant actuators in order to optimize a given performance index, following an *allocation policy*. What distinguishes *dynamic* control allocation from similar techniques is that, in the former, the allocation policy is a control law itself that is determined through a control design process and evaluated in real-time. Subsystems that are typically involved in a dynamic control allocation scheme are denoted as *dynamic allocators* or simply as *allocators*. The peculiarity of dynamic allocators is that they are designed as *add-ons* to a preexisting control system, *i.e.*, their design does not require a complete redesign of the control system, which could yield significant complications when optimization objectives are included in the problem. Instead, given the models of the plant and the control system, their design can be carried out *a posteriori* and they can subsequently be integrated into the control scheme as additional subsystems. The art of dynamic control allocation, therefore, lies in the design of these additional components with the goal in mind of not only optimizing the control action in terms of some criteria of practical relevance, but also of ensuring that the original control objectives are still met without any alteration of the desired output behavior.

Regarding TCV, a preliminary study has been carried out in [15], in which an early implementation of a dynamic coil current allocator has been integrated in the MATLAB code suite of TCV. Its performance has been validated by means of simulations on real shot data, which showed the feasibility of this approach. The rest of this chapter presents the work that has been carried out in this direction. First, a review of the current state of the art about dynamic control allocation is presented, together with a set of formulations that are of interest for the TCV case. Each of these has been considered, leading to some theoretical results that are illustrated in the following. Then, the design of a novel dynamic control allocator for TCV is presented, which is based on the latest advancements in the subject. Experimental results are discussed at the end of the chapter, showing the performance of the proposed novel allocator in a real scenario and constituting the first working implementation of its kind in the literature.

7.4 State of the art in dynamic control allocation

The evolution of technology over recent decades has led to the emergence of highly interconnected systems (see Chapters 1 and 2) characterized by increasingly complex mathematical models. A distinctive trait of many such systems is the presence of multiple actuation devices. These actuators, which may possess diverse capabilities and characteristics, may be kept as spare or operate in a *redundant* manner. In the latter case, the resulting controlled plants feature more actuators than sensors and measurement devices, leading to their classification as *over-actuated systems*. This configuration implies that an identical output behavior can be achieved through different input configurations. However, this flexibility is frequently perceived in practice as a liability, leading to control architectures that first reduce the model to a square plant through, *e.g.*, *squaring-down* techniques and then apply conventional control strategies. This approach not only foregoes the advantages of an intrinsically redundant plant structure but may also result in closed-loop systems with suboptimal performance or undesirable characteristics.

The ability to select control inputs that both align with prescribed closed-loop outputs and optimize specific performance criteria has therefore become crucial. These performance objectives may encompass enhanced system behavior, reduced power consumption, or improved reliability with respect to practical limitations. Given these compelling benefits, the allocation of control in input-redundant plants remains an active and vital research domain, as detailed in [16, 17, 18]. The growing significance of this field is evident in its diverse applications, spanning automotive systems [19], aeronautics [20], and tokamak fusion reactors [21, 22, 23, 15].

To properly contextualize relevant contributions, it is essential to distinguish between *static* and *dynamic* control allocation techniques. Static allocators simply redistribute control signals across plant inputs according to fixed laws expressed as, *e.g.*, gain matrices. Conversely, dynamic allocation consists in the implementation of allocation units as dynamic compensators within existing control schemes. The aim of these units is to preserve the original controlled plant output while optimizing the input or even the plant state evolutions through input redundancy. Foundational work in dynamic control allocation for linear systems includes [24, 25, 26, 27].

The notion of a dynamic control allocator is formalized in [24], where it stems from the concepts of *weak* and *strong input redundancy*, which characterize two possible plant configurations with respect to the effect of the allocation action on the output behavior. In particular, a *weakly input-redundant* plant is such that the allocation action becomes *invisible* in the output only at steady-state. Conversely, a *strongly input-redundant* plant rejects the input allocation action within the state evolution, making it invisible in the output at all times. This study is also the base for some notable work regarding the design of control allocators for tokamak plants [21, 22, 23, 15].

The design of a full-information regulator for over-actuated linear systems is considered in [25], introducing a geometric approach. This method is motivated by the geometric characterization of a feedback regulator and leads to significant developments concerning the effect of input redundancy on the *output invisibility* of the allocation action. The geometric approach is extended in [26, 27] accounting for model uncertainties. Moreover, the full-information regulator problem is addressed again in [28] where the selected approach consists in the parametrization of the solutions to the regulator equations, defining a space over which optimization methods are evaluated.

The notion of output invisibility is further explored in [29, 30], with the goal of optimizing the steady-state plant input without altering the steady-state plant output. The authors consider control systems subject to a sinusoidal reference input, which is also available to the allocation device. The main novelty of these works lies in the introduction of an architecture for the allocation device based on the interconnection of two subsystems: a *steady-state optimizer* and an *annihilator*. The former is responsible for the optimization of the steady-state plant input, while the latter ensures that the allocation action is invisible in the steady-state plant output. Their design is not decoupled but the optimizer can be designed after the annihilator. This architecture is particularly suitable for input-redundant plants with a weak input redundancy property and constitutes the starting point for the work presented in this chapter.

Furthermore, [31] analyzes the impact of plant model uncertainties on the design of a dynamic control allocator, while [32] proposes a functional approximation methodology. Static control allocation is evaluated in [33] against linear quadratic control, examining data requirements and achievable approximations. Alternative approaches include reinforcement learning-based dynamic allocation [34] and static optimal allocation achieved with a model predictive control design [35]. Dynamic control allocation also addresses implementation challenges like actuator saturation, as demonstrated in adaptive control contexts [36].

Some early results regarding control allocation for nonlinear systems are given in [37, 38]. In [37], the notion of weakly input-redundant plants is extended to nonlinear systems which admit a vector relative degree. This extension leads to the design of both static and dynamic input allocators for nonlinear systems that admit a special global normal form. Conversely, [38] proposes a Lyapunov-based approach to achieve optimal input allocation satisfying safety constraints, enforced by barrier functions.

7.5 Control requirements

A dynamic control allocator for the TCV shape controller is to be designed in a way that makes it easy to be integrated within the existing framework, not only in terms of its implementation but also of the modeling techniques on which it is based. For this work, the following formulations of a dynamic control allocation problem have been considered. Their evaluation has led to theoretical results that are illustrated in the following and allowed the author to identify the most suitable approach for the TCV case.

7.5.1 Preliminary definitions

Each dynamic system defined in the following is assumed to be *complete* unless differently specified. Consider a continuous-time, time-invariant dynamic system Σ described by the equations

$$\dot{\chi}(t) = f(\chi(t), r(t)) + g(\chi(t))v(t), \qquad t \in \mathbb{R}_{\geq 0}, \tag{7.1a}$$

$$\eta_p(t) = h_p(\chi(t), v(t)), \qquad t \in \mathbb{R}_{\geq 0}, \tag{7.1b}$$

$$\eta_o(t) = h_o(\chi(t)), \qquad t \in \mathbb{R}_{\geq 0}, \tag{7.1c}$$

where $\chi(t) \in \mathbb{R}^n$ is the state vector, $r(t) \in \mathbb{R}^{n_r}$ is the *reference* input, $v(t) \in \mathbb{R}^m$ is the *auxiliary* control input, and $\eta_p(t) \in \mathbb{R}^{q_p}$, $\eta_o(t) \in \mathbb{R}^{q_o}$ are the output vectors. Let the functions $f : R^n \times \mathbb{R}^{n_r} \to \mathbb{R}^n$, $g_i : \mathbb{R}^n \to \mathbb{R}^n$, $i = 1, \dots, m$, $h_p : \mathbb{R}^n \times \mathbb{R}^m \to \mathbb{R}^{q_o}$, and $h_o : \mathbb{R}^n \to \mathbb{R}^{q_o}$ be smooth. Σ is a *two-input two-output* (TITO) model that is convenient to deal with multiple kinds of real-world plants: the ones that are of interest here are those where Σ embeds a dynamic process and a given control system for it. These situations are further detailed in the following together with the peculiarities of each case that has been considered. η_p is an output that measures the overall performance of the system (namely, a *performance output*) and the output η_o follows a desired behavior (namely, a *regulated output*) usually by the action of some given component that is internal to Σ.

The reference input r is generated by the *exosystem* E described by the equations

$$\dot{w}(t) = \xi(w(t)), \qquad t \in \mathbb{R}_{\geq 0}, \tag{7.2a}$$

$$r(t) = \psi(w(t)), \qquad t \in \mathbb{R}_{\geq 0}, \tag{7.2b}$$

where $w(t) \in \mathbb{R}^{n_e}$ is the state vector, $w(0) = w_0$, and the functions $\xi : \mathbb{R}^{n_e} \to \mathbb{R}^{n_e}$ and $\psi : \mathbb{R}^{n_e} \to \mathbb{R}^{n_r}$ are smooth. The properties of E are characterized by the following "blanket assumptions", which implicitly hold from this point onward.

Assumption 7.1. *The origin $w = 0$ of $\mathbb{R}^{n_e}$ is a stable equilibrium point associated with ξ.*

Assumption 7.2. *There exists a compact and invariant set $\mathcal{W} \subset \mathbb{R}^{n_e}$ such that $w_0 \in \mathcal{W}^\circ$ with $\mathcal{W}^\circ \subseteq \mathrm{int}(\mathcal{W})$, $\mathcal{W}^\circ$ neighborhood of the origin of $\mathbb{R}^{n_e}$, and $w(t) \in \mathcal{W}$ for any $t \in \mathbb{R}_{\geq 0}$. Moreover, any $w_0 \in \mathcal{W}^\circ$ is such that the flow $\Phi_t^\xi(w_0)$ is defined for all $t \in \mathbb{R}$ and, for any neighborhood $\mathcal{U}^\circ$ of w_0 and for any real number $\tau > 0$, there exists a time $t_1 > \tau$ such that $\Phi_{t_1}^\xi(w_0) \in \mathcal{U}^\circ$, and a time $t_2 < -\tau$ such that $\Phi_{t_2}^\xi(w_0) \in \mathcal{U}^\circ$.*

In other words, Assumption 7.2 states that given $w_0 \in \mathcal{W}^\circ$, the trajectory $w(t)$ that originates in w_0 passes arbitrarily close to w_0 for $|t| \to +\infty$. This defines the point w_0 as *Poisson stable* (see [39]) and, since this holds for any $w_0 \in \mathcal{W}^\circ$, the system E is said to be Poisson stable. This property implies that no trajectory of (7.2) can decay to zero as time tends to infinity. The interconnection of the system Σ with the exosystem E is depicted in Figure 7.1.

The model described by the equations (7.1)-(7.2) is further characterized by the following assumptions. The first one is instrumental to enable the formulation of the control allocation problems and the design of dynamic allocators.

Assumption 7.3. *$m > q_o$.* ◦

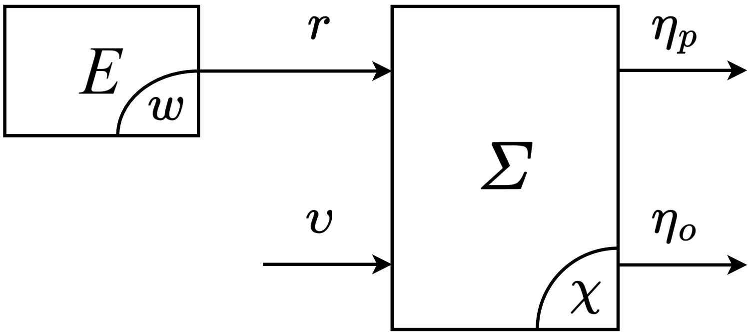

Figure 7.1: Block diagram of the system (7.1) interconnected to the exosystem (7.2).

Assumption 7.3 states that the system Σ is *over-actuated* with respect to the auxiliary input v and the regulated output η_o. The next assumption characterizes Σ as a preexisting dynamic system whose auxiliary input can be exploited to improve the performance indicator.

Assumption 7.4. *The system Σ is input-to-state stable (ISS) with respect to the input (r, v). Moreover, the equilibrium $\chi = 0$ of the system $\dot{\chi} = f(\chi, 0)$ is exponentially stable.* ∘

Finally, the following assumption clarifies the role of the exosystem E in the considered setting.

Assumption 7.5. *The exogenous input r, generated by the exosystem (7.2), is periodic of period T. Moreover, there exists a mapping $\pi : \mathcal{W}^\circ \to \mathbb{R}^n$ that, for any $w \in \mathcal{W}^\circ$, satisfies*

$$\frac{\partial \pi}{\partial w}\xi(w) = f(\pi(w), \psi(w)). \tag{7.3}$$

∘

It is worth to point out that Assumption 7.5 provides, for any $w_0 \in \mathcal{W}^\circ$, a *steady-state* *response* to the system (7.1)-(7.2) (see [39, Prop. 8.1.1]), denoted

$$\tilde{\chi}(t) = \chi(t, \pi(w_0), \psi(\Phi_t^{\xi}(w_0))), \quad t \in \mathbb{R}_{\geq 0}. \tag{7.4}$$

The performance and regulated outputs also admit steady-state responses denoted $\tilde{\eta}_p(\cdot)$ and $\tilde{\eta}_o(\cdot)$, respectively.

Let $\tau_0 \in \mathbb{R}_{\geq 0}$. The performance criterion with respect to which the behavior of Σ may be optimized is expressed by the following *Lagrange functional* of the form

$$J(v) = \int_{\tau_0}^{\tau_0+T} \ell(\tilde{\eta}_p(\tau))\, d\tau, \tag{7.5}$$

where the function $\ell : \mathbb{R}^{q_p} \to \mathbb{R}_{\geq 0}$ is smooth, convex, and radially unbounded.

The two dynamic control allocation problems that are of interest for this work are formalized as follows. The allocation action is implemented by means of the auxiliary input v of the system Σ and the nature of the following two problems originates from two different ways in which the effect of the auxiliary input on the regulated output can be evaluated.

Problem 7.1 (Dynamic allocation with steady-state output invisibility)**.** *Consider the system* (7.1) *and the exosystem* (7.2)*. Suppose that Assumptions 7.3, 7.4 and 7.5 hold. Find, if possible, an input allocation device (namely, an* allocator*) having input ϖ and output ς such that, when interconnected to Σ and E according to*

$$\varpi = r, \tag{7.6a}$$

$$\varsigma = v, \tag{7.6b}$$

ensures that the following requirements hold.

(P) Steady-state periodicity: *the allocator admits a limited steady-state output response, namely* $\tilde{\varsigma}(\cdot)$, *and it is periodic of period* T.

(I) Steady-state output invisibility: *the steady-state regulated output responses of the system given by eq.* (7.1) *both in presence and in absence of the allocation action, namely* $\tilde{\eta}_o^a(\cdot)$ *and* $\tilde{\eta}_o(\cdot)$, *respectively, are such that*

$$\tilde{\eta}_o^a(t) = \tilde{\eta}_o(t) \tag{7.7}$$

for all $t \in \mathbb{R}_{\geq 0}$.

(O) Steady-state optimality: *the interconnection of the allocator to* (7.1)-(7.2) *brings the cost functional* (7.5) *to its minimum.*

Problem 7.2 (Dynamic allocation with full output invisibility)**.** *Consider the system* (7.1) *and the exosystem* (7.2). *Suppose that Assumptions 7.3, 7.4 and 7.5 hold. Find, if possible, an input allocation device (namely, an* allocator*) having input* ϖ *and output* ς *such that, when interconnected to* Σ *and* E *according to*

$$\varpi = r, \tag{7.8a}$$

$$\varsigma = \upsilon, \tag{7.8b}$$

ensures that the following requirements hold.

(P) Steady-state periodicity: *the allocator admits a limited steady-state output response, namely* $\tilde{\varsigma}(\cdot)$, *and it is periodic of period* T.

(I) Full output invisibility: *the regulated output responses of the system given by eq.* (7.1) *both in presence and in absence of the allocation action, namely* $\eta_o^a(\cdot)$

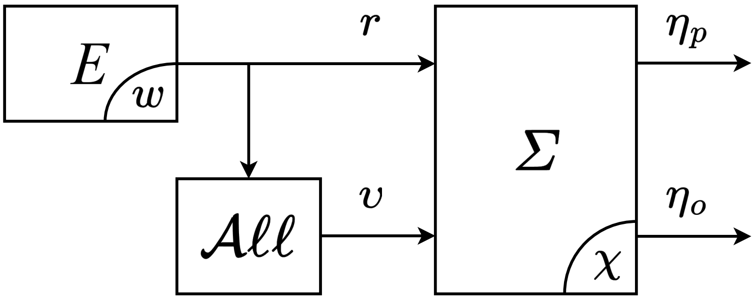

Figure 7.2: Block diagram of the system (7.1) interconnected to the exosystem (7.2) and to the input allocation device, enclosed in the $\mathscr{A}\ell\ell$ block.

and $\eta_o(\cdot)$*, respectively, are such that*

$$\eta_o^a(t) = \eta_o(t) \tag{7.9}$$

for all $t \in \mathbb{R}_{\geq 0}$.

(O) Steady-state optimality: *the interconnection of the allocator to* (7.1)-(7.2) *brings the cost functional* (7.5) *to its minimum.*

It is finally worth to point out that, in both problems, the optimization process is allowed to exhibit a transient behavior: what distinguishes the two situations is whether the effect of this transient behavior on the regulated output can be visible or not. The desired configuration is depicted in Figure 7.2.

7.6 Design of dynamic control allocators

The results that follow in the rest of this chapter provide solutions to Problems 7.1 and 7.2 based on different configurations of (7.1)-(7.2). The case of Problem 7.1 is examined considering both a linear and a nonlinear model for the system (7.1). On the other hand, Problem 7.2 is studied from the only point of view of linear systems but in multiple scenarios: when the model of Σ is fully known, when it is affected by parametric uncertainties, and when it includes sampled data devices.

7.6.1 Dynamic allocation with steady-state output invisibility

The case identified by Problem 7.1 where the systems (7.1)-(7.2) are linear has been studied in the literature and multiple solutions have been proposed, as reported in Section 7.4. However, none of these solutions retain their applicability when the model exhibits less convenient features, which is one of the reasons why the dynamic control allocation problem for nonlinear systems is still an open issue. This section presents the work that has been done in this direction, starting from the following motivating idea. Since, at its core, the dynamic control allocation problem is a constrained dynamic optimization problem, it is beneficial to reformulate it as such. This allows one to use approaches and tools belonging to the theory of optimal control, whose applicability also spans to nonlinear and broader classes of systems.

This approach has been tested initially on a linear feedback model of the form of Σ, interconnected to a linear exosystem E [40]. Consider a continuous-time, linear time-invariant dynamic system Θ of the form

$$\dot{x}(t) = A_p x(t) + B_p u(t), \qquad t \in \mathbb{R}_{\geq 0}, \tag{7.10a}$$

$$y(t) = C_p x(t), \qquad t \in \mathbb{R}_{\geq 0}, \tag{7.10b}$$

where $x(t) \in \mathbb{R}^{n_p}$ is the plant state vector, $u(t) \in \mathbb{R}^{m}$ is the control input, $y(t) \in \mathbb{R}^{q_o}$ is the plant output vector, and $A_p \in \mathbb{R}^{n_p \times n_p}$, $B_p \in \mathbb{R}^{n_p \times m}$, and $C_p \in \mathbb{R}^{q_o \times n_p}$ are constant matrices. In the considered setting, the system (7.10) is subject to the action of a continuous-time, linear time-invariant feedback controller Γ given by

$$\dot{\varphi}(t) = A_c\varphi(t) + B_c y(t) + P_c r(t), \qquad t \in \mathbb{R}_{\geq 0}, \tag{7.11a}$$

$$\rho(t) = C_c\varphi(t), \qquad t \in \mathbb{R}_{\geq 0}, \tag{7.11b}$$

where $\varphi(t) \in \mathbb{R}^{n_c}$ is the controller state vector and $\rho(t) \in \mathbb{R}^m$ is the controller output vector, and $A_c \in \mathbb{R}^{n_c \times n_c}$, $B_c \in \mathbb{R}^{n_c \times q_o}$, $P_c \in \mathbb{R}^{n_c \times n_r}$, and $C_c \in \mathbb{R}^{m \times n_c}$ are constant matrices. The interconnection of the plant Θ and the control system Γ forms the feedback loop Σ according to $u = \rho + v$, also yielding $n = n_p + n_c$; to keep the notation as concise as possible, until otherwise specified the plant output y is considered as the regulated output of the system Σ, *i.e.*, $\eta_o = y$. The following assumption characterizes the properties of the feedback interconnection.

Assumption 7.6. *The closed-loop feedback interconnection of* (7.10) *and* (7.11) *does not induce pole-zero cancellations, and it is well-posed and asymptotically stable.*

Let $s \in \mathbb{C}$. $W_{yu}(s)$ is the transfer matrix of the system (7.10) from the input u to the output y with $\rho = v = 0$, computed from the state-space description of Θ. $W_{yv}(s)$ is the transfer matrix of the closed loop Σ from the input v to the output y; moreover, let $(A_\Sigma, B_\Sigma, C_\Sigma, 0)$ be a realization of $W_{yv}(s)$. The dimensions of $W_{yu}(s)$ and $W_{yv}(s)$ are well determined and coincide with those of their respective state-space description thanks to Assumption 7.6.

The reference input in this setting is defined as the state of the exosystem E, which is a continuous-time linear time-invariant system of the form

$$\dot{w}(t) = Sw(t), \qquad t \in \mathbb{R}_{\geq 0}, \tag{7.12a}$$

$$r(t) = w(t), \qquad t \in \mathbb{R}_{\geq 0}, \tag{7.12b}$$

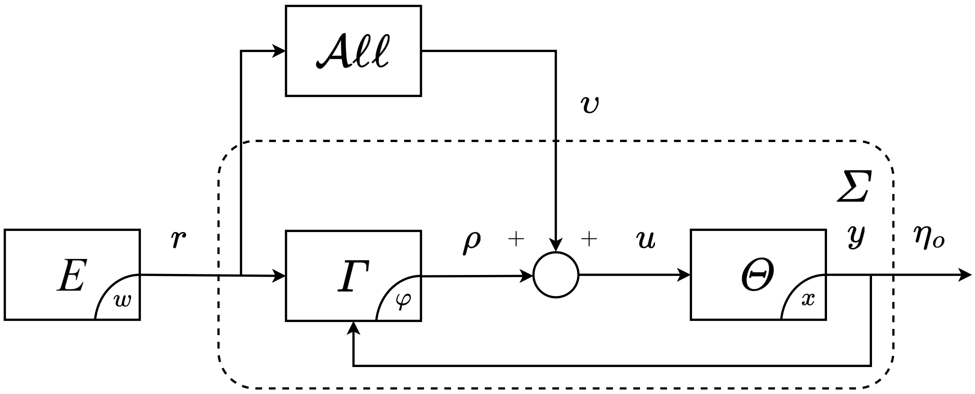

Figure 7.3: Block diagram of the feedback control system (7.10)-(7.11) interconnected to the exosystem (7.12); the input allocation device is enclosed in the $\mathscr{A}\ell\ell$ block.

where $S \in \mathbb{R}^{n_e \times n_e}$, $S = \mathrm{diag}(S_0, \dots, S_h)$ consists of the blocks defined as

$$\begin{aligned} S_0 &= 0, \\ S_i &= \begin{bmatrix} 0 & \omega_i \\ -\omega_i & 0 \end{bmatrix}, \quad i = 1, \dots, h, \end{aligned} \tag{7.13}$$

with $w(0) = w_0$. The frequencies $\{\omega_1, \dots, \omega_h\}$ in eq. (7.13) are such that $\omega_i = k_i \frac{2\pi}{T}$, $k_i \in \mathbb{N}$, $i = 1, \dots, h$.

The feedback controller (7.11), together with the exosystem (7.13), ensure that Assumptions 7.4 and 7.5 are satisfied in the setting provided by this formulation. The complete block diagram of the system Σ is shown in Figure 7.3.

Both the transfer matrices $W_{yu}(s)$ and $W_{yv}(s)$ admit left coprime polynomial factorizations (see [41, §7.8]) of the form

$$W_{yu}(s) = D_p^{-1}(s) N_p(s), \tag{7.14a}$$

$$W_{yv}(s) = D_\Sigma^{-1}(s) N_\Sigma(s), \tag{7.14b}$$

where $D_p(s)$, $D_\Sigma(s)$, $N_p(s)$ and $N_\Sigma(s)$ are polynomial matrices. The following assumption is introduced in this setting to simplify the discussion, although it implies that the controller (7.11) cannot embed an internal model of the exosystem (7.12).

Assumption 7.7. *The property*

$$\ker(N_\Sigma(0)) = \ker(N_p(0)), \tag{7.15a}$$

$$\ker(N_\Sigma(j\omega_i)) = \ker(N_p(j\omega_i)), \qquad i = 1,\dots,h, \tag{7.15b}$$

holds for any left coprime factorization (7.14). ∘

Such factorizations are not unique in general but it can be shown that Assumption 7.7 holds for one of them if and only if it holds for each one of them. The performance output in this case is defined as the plant input, *i.e.*, $\eta_p = u$. The effort demanded to the actuators is thus evaluated by the instantaneous cost function $\ell(\tilde{\eta}_p(t)) = \|\tilde{u}(t)\|_2^2$. Setting $\tau_0 = 0$ in eq. (7.5) yields the following formulation of the cost functional ℓ:

$$J(\tilde{u}) = \frac{1}{2}\int_0^T \|\tilde{u}(\tau)\|_2^2 \, d\tau. \tag{7.16}$$

Requirements (P), (I), and (O) of Problem 7.1 reveal the connection between *dynamic control allocation* on one hand and *constrained optimal control* on the other: the (P) item requires the asymptotic behavior to be T-periodic, the (O) item introduces the cost to optimize, and finally the (I) item describes the constraint to be satisfied to ensure that the optimal steady-state solution does not impact the original steady-state output performance induced by the controller (7.11). To substantiate this intuition, the definition of a few more items is required. Let $(A_\varphi, B_\varphi, C_\varphi, 0)$ be a state-space realization of the feedback loop Σ having w as input and ρ as output. Let $\Pi \in \mathbb{R}^{n\times n_e}$

be the solution of the Sylvester equation

$$\Pi S = A_\varphi \Pi + B_\varphi, \tag{7.17a}$$

and define $\bar{M} \in \mathbb{R}^{m \times n_e}$ as

$$\bar{M} = C_\varphi \Pi, \tag{7.17b}$$

termed *pseudo-steady-state output map.* Let $\tilde{\rho}(\cdot)$ denote the steady-state controller output. The following equality holds (see Appendix C, as well as [42, 43] for a proof):

$$\tilde{\rho}(t) = \bar{M} w(t), \quad t \in \mathbb{R}_{\geq 0}. \tag{7.18}$$

The finite-horizon optimal control problem corresponding to Problem 7.1 is formulated as follows.

Problem 7.3. *Consider the continuous-time feedback control system described by* (7.10), (7.11)*, and the exosystem* (7.12)*. Suppose that Assumptions 7.3, 7.4, 7.5, 7.6 and 7.7 hold, and define* $\bar{M}$ *as in* (7.17)*. Determine, if it exists, a function* $v^\star(\cdot) \in \mathscr{L}_2([0,T])$ *that*

(i) minimizes the cost functional

$$J(v) = \frac{1}{2}\int_0^T \left\| \bar{M} w(\tau) + v(\tau) \right\|_2^2 \, d\tau; \tag{7.19}$$

(ii) satisfies the integral constraint

$$\lim_{t_0 \to -\infty} \int_{t_0}^{t} C_\Sigma e^{A_\Sigma (t-\tau)} B_\Sigma \hat{v}(\tau) \, d\tau = 0, \quad t \in \mathbb{R}_{\geq 0}, \tag{7.20}$$

where $\hat{v} : \mathbb{R} \to \mathbb{R}^m$ *denotes the periodic extension of* v*, namely* $\hat{v}(t) = v(\mathrm{mod}(t,T))$*,* $t \in \mathbb{R}_{\geq 0}$.

It is worth pointing out that item (ii) of Problem 7.3 constitutes a somewhat novel class of constraints for optimal control problems. In particular, since the left-hand side of (7.20) represents the steady-state output response of the closed loop identified by $(A_\Sigma, B_\Sigma, C_\Sigma, 0)$ with $v = v$, item (I) of Problem 7.1 is equivalent to the constraint (7.20) of Problem 7.3.

Remark 7.1. *The relation between Problem 7.1 and Problem 7.3 establishes the basis for a deeper understanding of both. Having determined the solution $v^\star(\cdot)$ of Problem 7.3, it is possible to satisfy the requirements of Problem 7.1 simply by injecting $v(t) = v^\star(\mathrm{mod}(t,T))$ into the closed loop; the linearity and asymptotic stability of Σ imply that the resulting closed loop converges to a steady-state satisfying the requirements of Problem 7.1. Note that while an allocated closed loop using a generic allocator solving Problem 7.1 generally exhibits transients in both $v(\cdot)$, which is computed asymptotically, and in the overall system response, the proposed approach directly generates the steady-state evolution of $v(\cdot)$, although the remaining signals in the overall system response still exhibit transients before converging to their allocated evolution.*

It is possible to derive a closed-form solution to Problem 7.3. However, some preliminary technical results are required. For clarity, introduce the notation

$$\zeta(t) = \bar{M} w(t), \quad t \in \mathbb{R}_{\geq 0}. \tag{7.21}$$

By standard results about Fourier series expansion of functions in $\mathscr{L}_2([0,T])$, it is possible to write

$$\zeta(t) = \zeta_0 + \sum_{k=1}^{+\infty} (a_k \sin(k\omega t) + b_k \cos(k\omega t)), \qquad t \in \mathbb{R}_{\geq 0}, \tag{7.22a}$$

$$v^\star(t) = v_0^\star + \sum_{k=1}^{+\infty} (\alpha_k \sin(k\omega t) + \beta_k \cos(k\omega t)), \qquad t \in \mathbb{R}_{\geq 0}, \tag{7.22b}$$

where $\zeta_0, v_0^\star, a_k, b_k, \alpha_k, \beta_k \in \mathbb{R}^m$. Moreover, by the properties of (7.2) and (7.12), only a finite number of coefficients a_k, b_k are non-zero in (7.22a); the same property holds for (7.22b) as shown in the following. The first Lemma, considering a single frequency component of (7.22b), characterizes the relation imposed by the constraint (7.20) on its coefficients.

Lemma 7.1. *Let $\bar\omega = 2\pi\bar k/T$, with $\bar k \in \mathbb{N}\cup\{0\}$, and*

$$v(t) = \alpha\sin(\bar\omega t) + \beta\cos(\bar\omega t), \quad t \in \mathbb{R}_{\geq 0}, \tag{7.23}$$

where $\alpha, \beta \in \mathbb{R}^m$, and let $\vartheta := [\alpha'\ \beta']'$, and define the matrix

$$\Psi(\bar\omega) := \begin{bmatrix} \mathrm{Im}(N_\Sigma(j\bar\omega)) & \mathrm{Re}(N_\Sigma(j\bar\omega)) \\ -\mathrm{Re}(N_\Sigma(j\bar\omega)) & \mathrm{Im}(N_\Sigma(j\bar\omega)) \end{bmatrix}. \tag{7.24}$$

Then, (7.20) *holds for $v(\cdot)$ in* (7.23) *if and only if either one of the following conditions holds:*

$$\Psi(\bar\omega)\vartheta = 0, \tag{7.25a}$$

$$\vartheta = \Psi^\perp(\bar\omega)z,\ z \in \mathbb{R}^\nu, \tag{7.25b}$$

where $\nu = \dim(\ker(\Psi(\bar\omega)))$. ∘

Proof. Rewrite

$$v(t) = \alpha\sin(\bar\omega t) + \beta\cos(\bar\omega t) = \frac{1}{2}\left(e^{j\bar\omega t}(\beta - j\alpha) + e^{-j\bar\omega t}(\beta + j\alpha)\right). \tag{7.26}$$

By linearity and Assumption 7.6, under the input $v(\cdot)$ the steady-state response of Σ in the left hand side of (7.20) is given by

$$\tilde{y}(t) = \frac{1}{2}e^{j\bar{\omega}t}W_{yv}(j\bar{\omega})(\beta - j\alpha) + \frac{1}{2}e^{-j\bar{\omega}t}W_{yv}(-j\bar{\omega})(\beta + j\alpha). \tag{7.27}$$

Considering that the two addends in the last expression are each the complex conjugate of the other, it follows that $\tilde{y}(\cdot)$ is zero if and only if

$$W_{yv}(j\bar{\omega})(\beta - j\alpha) = 0;$$

in turn, considering (7.14) and that Assumption 7.6 implies that $D_\Sigma(j\bar{\omega})$ is invertible, the last equation is equivalent to

$$N_\Sigma(j\bar{\omega})(\beta - j\alpha) = 0. \tag{7.28}$$

Imposing that both the imaginary part and the real part of $N_\Sigma(j\bar{\omega})(\beta - j\alpha)$ must be zero, (7.25a) follows. It is worth to point out that the case $\bar{k} = 0$, *i.e.*, $\bar{\omega} = 0$, is also included by (7.25a) since in this case $\alpha = 0$, $\beta = v_0^\star$, $\mathrm{Im}(N_\Sigma(0)) = 0$, $\mathrm{Re}(N_\Sigma(0)) = N_\Sigma(0)$, thus (7.25a) reduces to an identity ($0 = 0$) plus the condition $N_\Sigma(0)v_0^\star = 0$. □

The property stated by Lemma 7.1 parametrizes the functions $\hat{v}^\star(\cdot)$ satisfying (7.20), thus allowing one to translate the problem of optimizing (7.19) under the constraint (7.20) into an unconstrained optimization problem on a suitably defined, smaller domain. Focusing again on the case of a single frequency component, the next Lemma shows how the coefficients of $\zeta(\cdot)$ and $v^\star(\cdot)$ are related when the cost (7.19) is also minimized.

Lemma 7.2. *Let* $\bar{\omega} = 2\pi\bar{k}/T$, *with* $\bar{k} \in \mathbb{N} \cup \{0\}$, *and*

$$\zeta(t) = a\sin(\bar{\omega}t) + b\cos(\bar{\omega}t), \qquad t \in \mathbb{R}_{\geq 0}, \tag{7.29a}$$

$$v(t) = \alpha\sin(\bar{\omega}t) + \beta\cos(\bar{\omega}t), \qquad t \in \mathbb{R}_{\geq 0}, \tag{7.29b}$$

where $a, b, \alpha, \beta \in \mathbb{R}^m$, $\theta := [a'\ b']'$, $\vartheta := [\alpha'\ \beta']'$.

Then, $v(\cdot)$ *minimizes* (7.19) *under the constraint* (7.20) *if and only if* $\vartheta = \vartheta^\star(\bar{\omega})$, *with*

$$\vartheta^\star(\bar{\omega}) = -\Psi^\perp(\bar{\omega})(\Psi^\perp(\bar{\omega}))^\dagger\theta, \tag{7.30}$$

and $\vartheta^\star(\bar{\omega}) = [\alpha^\star(\bar{\omega})'\ \beta^\star(\bar{\omega})']'$. ∘

Proof. The computation of the cost (7.19) with (7.29) reduces to

$$\begin{aligned} &\frac{1}{2}\int_0^T \|\zeta(\tau) + v(\tau)\|_2^2\, d\tau = \\ &= \frac{1}{2}\int_0^T \left\|(a+\alpha)\sin(\bar{\omega}\tau) + (b+\beta)\cos(\bar{\omega}\tau)\right\|_2^2 d\tau = \\ &= \frac{1}{4}\left\|\begin{bmatrix}(a+\alpha)\\(b+\beta)\end{bmatrix}\right\|_2^2, \end{aligned} \tag{7.31}$$

where the integral over a period of the mixed term $\sin(\bar{\omega}\tau)\cos(\bar{\omega}\tau)$ evaluates to zero. From Lemma 7.1, in order to satisfy the constraint (7.20) the vector ϑ must be of the form (7.25b), namely $\vartheta = \Psi^\perp(\bar{\omega})z$ with $z \in \mathbb{R}^\nu$. Then, the optimal coefficients $\vartheta^\star(\bar{\omega}) = [\alpha^\star(\bar{\omega})'\ \beta^\star(\bar{\omega})']'$ are obtained by finding $z^\star \in \mathbb{R}^\nu$ such that

$$z^\star = \arg\min_{z \in \mathbb{R}^\nu} \bar{J}(z), \tag{7.32}$$

where

$$\bar{J}(z) = \frac{1}{2}\left\|\theta + \Psi^{\perp}(\bar{\omega})z\right\|_2^2. \tag{7.33}$$

Note that (7.32) is an unconstrained, finite-dimensional optimization problem completely equivalent to Problem 7.3 in the setting defined by the statement of this Lemma. Thus, the solution may be computed by zeroing the gradient of the cost function, namely,

$$\frac{\partial \bar{J}(z)}{\partial z} = \left(\theta + \Psi^{\perp}(\bar{\omega})z\right)' \Psi^{\perp}(\bar{\omega}) = 0, \tag{7.34}$$

yielding

$$\left(\Psi^{\perp}(\bar{\omega})\right)'\theta + \left(\Psi^{\perp}(\bar{\omega})\right)'\Psi^{\perp}(\bar{\omega})z = 0, \tag{7.35}$$

$$z^{\star} = -\left(\Psi^{\perp}(\bar{\omega})\right)^{\dagger}\theta, \tag{7.36}$$

and then (7.30) is given by (7.25b). Finally, this solution yields the unique global minimum for (7.32) and consequently the solution for Problem 7.3 by the strict convexity of $\bar{J}$:

$$\frac{\partial^2 \bar{J}(z)}{\partial z^2} = \Psi^{\perp}(\bar{\omega})'\Psi^{\perp}(\bar{\omega}) > 0 \tag{7.37}$$

holds by construction, since $\Psi^{\perp}(\bar{\omega})$ is full column rank. □

Remark 7.2. *Consider* (7.29a) *and recall* (7.21)*, namely the fact that* $\zeta(t) = \bar{M}w(t)$ *with* $\bar{M}$ *given by* (7.17)*. The two expressions are reconciled noting that if*

$$w(t) = a_w \sin(\bar{\omega}t) + b_w \cos(\bar{\omega}t), \quad t \in \mathbb{R}_{\geq 0}, \tag{7.38}$$

with $a_w, b_w \in \mathbb{R}^{n_e}$*, then* $\theta = [a'\ b']' = \bar{M}[a_w'\ b_w']'$ *holds.* ▲

A straightforward Corollary of the previous result is deduced from (7.30). It tells, in essence, that if $\zeta(\cdot)$ contains a single harmonic with zero amplitude, then the optimal $v(\cdot)$ with the same single harmonic has zero amplitude as well.

Corollary 7.1. *If* $v(\cdot)$, $\zeta(\cdot)$ *are given by* (7.29), *with* $v(\cdot)$ *minimizing* (7.19) *under the constraint* (7.20) *and* $a = b = 0$ *in* (7.29a), *then* $\alpha = \beta = 0$ *in* (7.29b). $\circ$

The following result exploits the previous Lemmas and Corollary to describe the optimal solution in the general case where $\zeta(\cdot)$ and $v(\cdot)$ are given by (7.22).

Lemma 7.3. *Let* (7.21), (7.22) *hold. Let* $v^\star(\cdot)$ *be the minimizer of* (7.19) *under the constraint* (7.20) *and the excitation* $\zeta(\cdot)$. *Then,* $v^\star(\cdot)$ *contains all and only the frequency components present, with nonzero coefficients, in* $\zeta(\cdot)$, *and the coefficients* $\theta_k := [a_k' \; b_k']'$, $\vartheta_k := [\alpha_k' \; \beta_k']'$ *are related by* (7.30) *to* $\bar{\omega} = k\omega$. $\circ$

Proof. The proof simply consists of showing that the minimization of the cost in the case of several frequency components reduces to the separate minimization of each frequency component so that the previous results immediately apply. To avoid cumbersome notation, let

$$A_k = a_k + \alpha_k, B_k = b_k + \beta_k, \qquad k = 1, \dots \tag{7.39}$$

which enable to write the instantaneous cost in (7.19) as

$$\begin{aligned} &\frac{1}{2}\|\zeta(t) + v(t)\|_2^2 = \frac{1}{2}\left\|\gamma_0 + \delta(t) + \epsilon(t)\right\|_2^2 \\ &= \frac{1}{2}(\gamma_0'\gamma_0 + \delta(t)'\delta(t) + \epsilon(t)'\epsilon(t)) + \gamma_0'\delta(t) + \delta(t)'\epsilon(t) + \gamma_0'\epsilon(t), \end{aligned} \tag{7.40}$$

with $t \in \mathbb{R}_{\geq 0}$, and the vector γ_0 and the functions $\delta(\cdot)$ and $\epsilon(\cdot)$ defined as

$$
\begin{aligned}
\gamma_0 &= \zeta_0 + v_0^\star, \\
\delta(t) &= \sum_{k=1}^{+\infty} A_k \sin(k\omega t), \quad \epsilon(t) = \sum_{k=1}^{+\infty} B_k \cos(k\omega t).
\end{aligned}
\tag{7.41}
$$

The functional to be minimized is

$$
\begin{aligned}
&\int_0^T \left(\left\|\gamma_0\right\|_2^2 + \|\delta(t)\|_2^2 + \|\epsilon(t)\|_2^2\right) dt \\
&= T\left\|\gamma_0\right\|_2^2 + \sum_{k=1}^{+\infty}\left(\|A_k\|_2^2 \int_0^T \sin^2(k\omega t)\, dt\right) + \sum_{k=1}^{+\infty}\left(\|B_k\|_2^2 \int_0^T \cos^2(k\omega t)\, dt\right) \\
&= T(\zeta_0 + v_0^\star)'(\zeta_0 + v_0^\star) + \frac{\pi}{\omega}\sum_{k=1}^{+\infty}\left[(a_k + \alpha_k)'(a_k + \alpha_k) + (b_k + \beta_k)'(b_k + \beta_k)\right],
\end{aligned}
\tag{7.42}
$$

which follows considering the expansion of (7.40), noting that mixed terms contain products involving sine and cosine functions that, when integrated over a period, sum to zero, and recalling (7.39). On inspection, it is clear that the overall cost is simply the sum of the costs of the single frequency components; these costs are minimized by the coefficient determined in the previous Lemmas. □

The following result is thus obtained collecting the intermediate ones provided by the previous Lemmas.

Theorem 7.1. *Consider the feedback control system described by* (7.10)-(7.12) *and suppose that Assumptions 7.3, 7.4, 7.5, 7.6 and 7.7 hold. Then, there exists a unique solution $v^\star$ to Problem 7.3, of the form*

$$
v^\star(t) = v_0^\star + \sum_{i=1}^{h} \alpha_i^\star \sin(\omega_i t) + \beta_i^\star \cos(\omega_i t), \quad t \in \mathbb{R}_{\geq 0},
\tag{7.43}
$$

with

$$v_0^\star = -N_\Sigma^\perp(0)(N_\Sigma^\perp(0))^\dagger \zeta_0, \tag{7.44}$$

$$\begin{bmatrix} \alpha_i^\star \\ \beta_i^\star \end{bmatrix} = -\Psi^\perp(\omega_i)(\Psi^\perp(\omega_i))^\dagger \begin{bmatrix} a_i \\ b_i \end{bmatrix}. \tag{7.45}$$

○

The results illustrated so far have been corroborated by means of some numerical simulations reported below. The following numerical model has been adopted for the plant (7.10):

$$A = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & -2 & 0 & 0 \\ 0 & 0 & 0 & 3 \\ 0 & -4 & 1 & 0 \end{bmatrix}, \ B = \begin{bmatrix} 1 & 0 & -1 \\ 1 & 0 & 1 \\ -1 & -1 & 0 \\ 0 & -1 & 1 \end{bmatrix}, \ C = \begin{bmatrix} 1 & 0 & 1 & 0 \end{bmatrix}. \tag{7.46}$$

The matrices describing the stabilizing controller (7.11) have been obtained by an LQG design with identity weights, restricting it to only the first two inputs of the plant (7.46). This choice has been made to better highlight the optimization action of the allocation device, which is instead allowed to modulate the control action among all the three inputs of (7.46). The exosystem is defined as in (7.12) for the set of frequencies given by $\{\omega_1 = 5,\ \omega_2 = 10,\ \omega_3 = 15\}$, while its initial condition has been chosen equal to $w_0 = [2\ 0\ 4\ 0\ 6\ 0]'$. Finally, it is assumed that the initial conditions for both the plant and the controller are equal to zero, and all simulations are carried out by setting the final time equal to $t_f = 7\ s$.

The purpose of the proposed simulations is to compare the performance of a state-of-the-art hybrid allocator, designed as in [29], against one that is implemented by

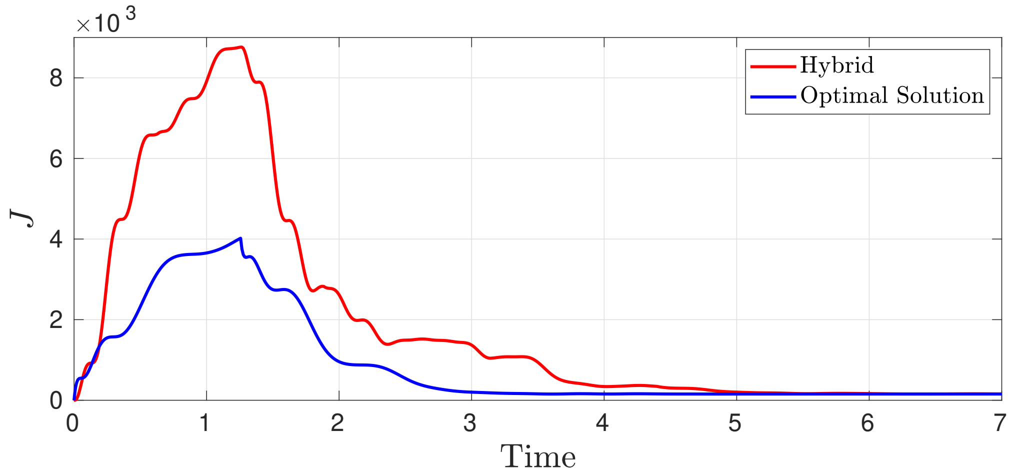

Figure 7.4: Cost in the presence of input allocation by means of a hybrid allocator (red), and of the periodic extension of the optimal solution to Problem 7.3 (blue).

applying the results presented above. The hybrid allocator is based on a gradient descent algorithm with the gradient decay rate γ selected as $\gamma = 2$. The implementation of the proposed allocator, based on the previous results, is based on the approach described in Remark 7.1, namely applying a periodic extension of the optimal solution $v^\star$ to Problem 7.3. Figure 7.4 reports the time histories of the costs for the considered cases, showing that both solutions converge to the same steady-state cost, which is equal to $J(v^\star) = 155.836$. The behaviors of the differences between the outputs of the non-allocated and allocated closed loops for the proposed approach and the state of the art are depicted in Figure 7.5, where the convergence to zero of the considered signals proves the satisfaction of the steady-state output invisibility requirement. The outputs of the state-of-the-art hybrid allocator are shown in Figure 7.6 against the components of $v^\star(t)$, showing that both signals become coincident after a transient phase.

This study provides a first step towards the integration of optimal control theory tools and results within dynamic control allocation settings. The proposed approach allows the designer to build a control allocation device which acts, in essence, as a

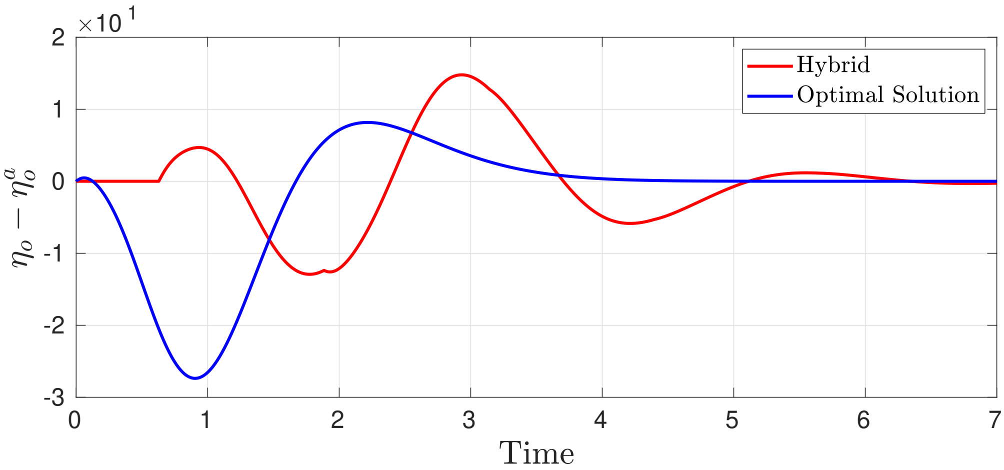

Figure 7.5: Differences between the outputs of the non-allocated and allocated closed loops, in the presence of a hybrid allocator (red) and of the periodic extension of the optimal solution to Problem 7.3 (blue).

periodic function generator whose dynamic model is entirely determined in terms of its frequency components by Theorem 7.1. Conversely, the previous state-of-the-art solution is described by more complex hybrid dynamics.

To generalize the approach validated by Theorem 7.1, it is essential to recall the original, nonlinear model. Thus, consider the dynamical system (7.1), subject to the reference input produced by the exosystem (7.2). Suppose that Assumptions 7.3, 7.4 and 7.5 hold and consider the steady-state cost functional (7.5), which evaluates the steady-state performance output of the TITO model Σ subject to the allocation action of the auxiliary input v. Similarly to the previous linear case, let $\ell(\tilde{\eta}_p(t)) = \|\tilde{\eta}_p(t)\|_2^2$ and $\tau_0 = 0$ in (7.5), yielding

$$J(v) = \frac{1}{2}\int_0^T \|\tilde{\eta}_p(\tau)\|_2^2 \, d\tau. \tag{7.47}$$

In this scenario, as in many others reported in the literature (see Section 7.4), the most complex requirement set by Problem 7.1 is the output invisibility item (I); in the linear case of Problem 7.3 it was still included as a problem constraint. Conversely,

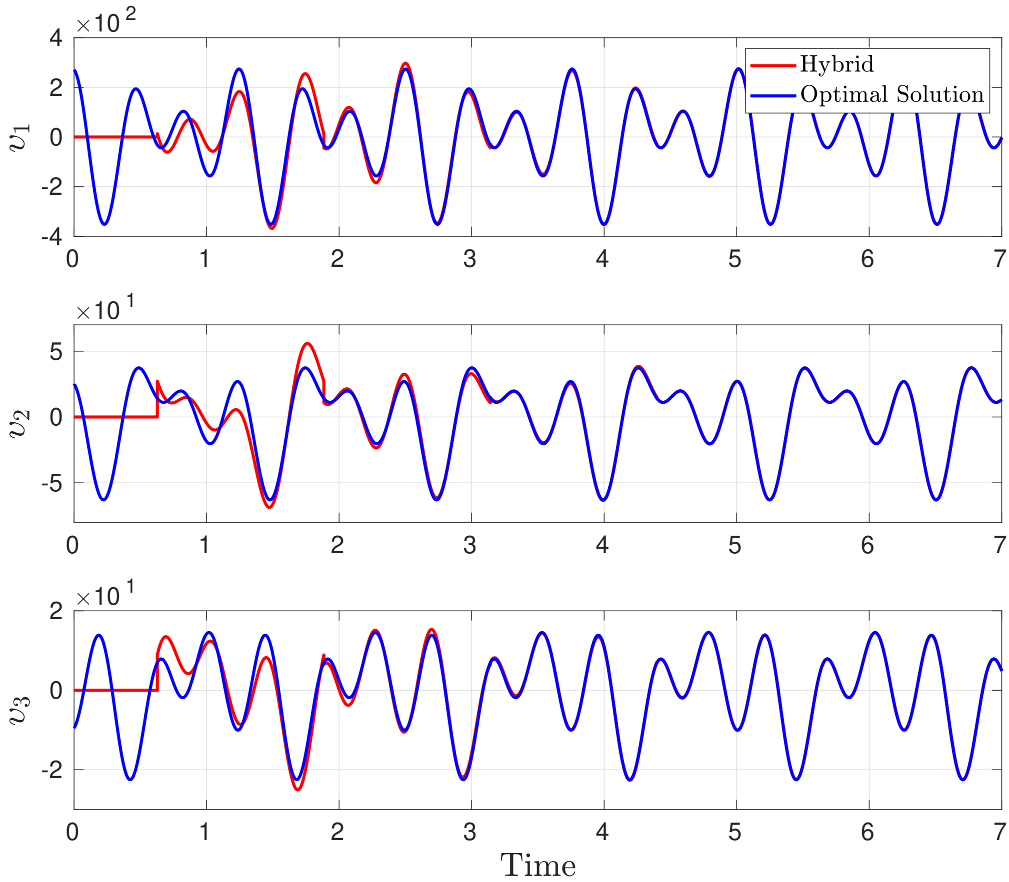

Figure 7.6: Outputs of a hybrid allocator (red) versus the components of the periodic extension of the optimal solution to Problem 7.3 (blue).

the approach hereby proposed for the original formulation of Problem 7.1 involving (7.1)-(7.2) takes a different path: Problem 7.1 is reformulated as an optimal control problem that approximates the original one by the use of *soft constraints* for output invisibility [44]. Then, a numerical algorithm is developed to determine candidate solutions to the approximated problem.

To substantiate this intuition a few preliminary definitions are in order. Let $\chi_n(t)$, $\chi_a(t) \in \mathbb{R}^n$ denote the state evolution of (7.1) in the absence or in the presence of the allocator, respectively, that is

$$\dot{\chi}_n(t) = f(\chi_n(t), r(t)), \qquad t \in \mathbb{R}_{\geq 0}, \tag{7.48a}$$

$$\dot{\chi}_a(t) = f(\chi_a(t), r(t)) + g(\chi_a(t))v(t), \qquad t \in \mathbb{R}_{\geq 0}, \tag{7.48b}$$

and let $\tilde{\chi}_n(\cdot)$, $\tilde{\chi}_a(\cdot)$ denote the corresponding steady-state responses, provided the latter exists. Define the *output invisibility function* $h_i(\chi_n, \chi_a) : \mathbb{R}^n \times \mathbb{R}^n \to \mathbb{R}$ such that

$$h_i(\chi_n(t), \chi_a(t)) = \frac{1}{2} \left\| h_o(\chi_n(t)) - h_o(\chi_a(t)) \right\|_2^2, \quad t \in \mathbb{R}_{\geq 0}. \tag{7.49}$$

A constrained optimal control problem that approximates Problem 7.1 involving soft constraints is formulated as follows.

Problem 7.4. *Consider the system* (7.1) *and suppose that Assumptions 7.3, 7.4 and 7.5 hold. Fix* $w_0 \in \mathscr{W}^\circ$, $w(0) = w_0$. *Let* $h_i(\chi_n(t), \chi_a(t))$ *be defined as in* (7.49). *Let* $v : \mathbb{R} \to \mathbb{R}^m$ *and define the cost functionals*

$$J_p(v) = \frac{1}{2} \int_0^T \left\| h_p(\tilde{\chi}_a(t), v(t)) \right\|_2^2 dt, \tag{7.50a}$$

$$J_i(v) = \int_0^T h_i(\tilde{\chi}_n(t), \tilde{\chi}_a(t))\, dt. \tag{7.50b}$$

Determine, if it exists, a function $v^\star(\cdot) \in \mathscr{L}_2([0, T])$, *periodic of period* T, *such that the cost functional*

$$J(v) = J_p(v) + \alpha J_i(v), \tag{7.51}$$

with $\alpha \in \mathbb{R}_{>0}$, *is minimized, satisfying the differential constraints* (7.48a)-(7.48b) *with* $v = v$.

Remark 7.3. *The output invisibility requirement,* i.e., *item (I) of Problem 7.1, represents the main challenge of dynamic input allocation problems in general. In Problem 7.4, by*

(7.50b) *and* (7.51)*, this requirement is included as part of the minimization objective, implying that the solutions to Problem 7.4 can only approximate the behavior of those to Problem 7.1. The quality of the approximation depends on the specific solutions to Problem 7.4 that can be determined, using numerical methods, as* $\alpha \to +\infty$. ▲

Remark 7.4. *While a generic allocator that solves Problem 7.1 may generally exhibits transients in its output* ς*, the solution to Problem 7.4 consists of a steady-state behavior repeated over the time interval* $[0, T]$*, similarly to the previous linear case [40]. However, there may still be transients in the overall system response of* Σ *and the remaining signals as they converge to their steady-state, allocated behavior.* ▲

Let $\lambda_n(t), \lambda_a(t) \in \mathbb{R}^n$ be the costate variables, associated to the nominal (*i.e.*, non-allocated) and allocated state variables, respectively. In the following, time dependencies are omitted to ease the notation. Recalling equations (7.48) to (7.51), define the Hamiltonian function

$$\begin{aligned} &H(\chi_n(t), \chi_a(t), \lambda_n(t), \lambda_a(t), v(t)) \\ &= \frac{1}{2}\left\|h_p(\chi_a(t), v(t))\right\|_2^2 + \alpha h_i(\chi_n(t), \chi_a(t)) + \lambda_n(t)' f(\chi_n(t), r(t)) \\ &+ \lambda_a(t)'(f(\chi_a(t), r(t)) + g(\chi_a(t)) v(t)), \quad t \in \mathbb{R}_{\geq 0}. \end{aligned} \tag{7.52}$$

To simplify the notation, define

$$\begin{aligned} H_v(\chi_a(t), \lambda_a(t), v(t)) &= \frac{\partial H(\chi_n(t), \chi_a(t), \lambda_n(t), \lambda_a(t), v(t))}{\partial v} \\ &= h_p(\chi_a(t), v(t))' \frac{\partial h_p(\chi_a(t), v(t))}{\partial v} + \lambda_a(t)' g(\chi_a(t)), \quad t \in \mathbb{R}_{\geq 0}, \end{aligned} \tag{7.53}$$

and note that the costate dynamics are given by

$$\begin{aligned}
\dot{\lambda}_n(t) &= -\alpha\frac{\partial h_i(\chi_n(t),\chi_a(t))'}{\partial \chi_n} - \frac{\partial f(\chi_n(t),r(t))'}{\partial \chi_n}\lambda_n(t), && t\in\mathbb{R}_{\geq 0}, && (7.54a)\\
\dot{\lambda}_a(t) &= -\frac{\partial h_p(\chi_a(t),v(t))'}{\partial \chi_a}h_p(\chi_a(t),v(t))\\
&\quad -\alpha\frac{\partial h_i(\chi_n(t),\chi_a(t))'}{\partial \chi_a} - \frac{\partial f(\chi_a(t),r(t))'}{\partial \chi_a}\lambda_a(t) && t\in\mathbb{R}_{\geq 0}, && (7.54b)\\
&\quad -\sum_{\iota=1}^{m}\left(v_\iota(t)\frac{\partial g_\iota(\chi_a(t))'}{\partial \chi_a}\right)\lambda_a(t),
\end{aligned}$$

where $v_\iota(t)$ and $g_\iota(t)$ denote the ι-th component and column of $v(t)$ and $g(t)$, respectively. Given $u(t)\in\mathbb{R}^m$, denote again with $\hat{u}:\mathbb{R}\to\mathbb{R}^m$ the periodic extension of $u(t)$, *i.e.*, $\hat{u}(t)=u(\mathrm{mod}(t,T))$. Let $\beta,\gamma\in\mathbb{R}_{>0}$, $\hat{v}(t)\in\mathbb{R}^m$, and $\chi_n(t)\in\mathbb{R}^n$ be the solution of (7.48a). Pick a final time $t_f=aT$ with $a\in\mathbb{N}$ large enough to include a reasonable number of periods, an arbitrary $v_0(t)\in\mathbb{R}^m$, let $(\lambda_n(t_f),\lambda_a(t_f))=(0,0)$, and define

$$\delta(H_v(\chi_a(t),\lambda_a(t),v(t)))=\left(\int_{t_f-T}^{t_f}H_v(\chi_a(t),\lambda_a(t),v(t))H_v(\chi_a(t),\lambda_a(t),v(t))'\,dt\right)^{1/2}. \tag{7.55}$$

The proposed method is summarized in Algorithm 1, and its properties are characterized by the following statement.

Theorem 7.2. *Consider the system* (7.1), *fix* $\alpha>0$, *and suppose that Assumptions 7.3, 7.4 and 7.5 hold. Then, given* $v_0:\mathbb{R}\to\mathbb{R}^m$, *and* $t_f=aT$, *there exists a sequence* $\{\gamma^{(k)}\}\subset\mathbb{R}_{>0}$ *such that the sequence of successive updates* $\{\tilde{v}^{(k)}(t)\}$, $t\in\mathbb{R}_{\geq 0}$, *produced by Algorithm 1 converges to an extremal solution of Problem 7.4. Furthermore, if Problem 7.4 admits an optimal solution* $v^\star$ *which is the unique extremal, then*

$$\lim_{k\to+\infty}\left\|\tilde{v}^{(k)}(t)-v^\star(t)\right\|=0,\quad t\in\mathbb{R}_{\geq 0}. \tag{7.56}$$

∘

Algorithm 1 Steady-state steepest descent algorithm [44].

Input: $\beta, \gamma, T, t_f, v_0(t), \chi_n(t)$
Output: $\tilde{v}(t)$

let $t \in [0, t_f]$
$\tilde{v}^{(0)}(t) \leftarrow v_0(t)$
for $k = 1, \ldots$ **do**
  $\chi_a^{(k)}(t) \leftarrow$ solution of (7.48b) with $v(t) = \tilde{v}^{(k-1)}(t)$
  $(\lambda_n^{(k)}(t), \lambda_a^{(k)}(t)) \leftarrow$ backward integration of (7.54) with $(\chi_a(t), v(t)) = (\chi_a^{(k)}(t), \tilde{v}^{(k-1)}(t))$
  if $\delta(H_v(\chi_a^{(k)}(t), \lambda_a^{(k)}(t), \tilde{v}^{(k-1)}(t))) < \beta$ **then return** $\tilde{v}(t) \leftarrow \tilde{v}^{(k-1)}(t)$
  else
    $u^{(k)}(t) \leftarrow \tilde{v}^{(k-1)}(t) - \gamma H_v(\chi_a^{(k)}(t), \lambda_a^{(k)}(t), \tilde{v}^{(k-1)}(t))$
    $u_p^{(k)}(\tau) \leftarrow u^{(k)}(t_f - T + \tau)$, with $\tau \in [0, T]$
    $\tilde{v}^{(k)}(t) \leftarrow \hat{u}_p^{(k)}(t)$ ▷ Periodic extension.
  end if
end for

Proof. The existence of the sequence $\{\gamma^{(k)}\}$, as well as its convergence properties, are guaranteed by the Steepest Descent Algorithm (see, *e.g.*, [45, Sec. 6.2]) adapted in Algorithm 1 to the steady-state optimal control problem formalized in Problem 7.4. □

Remark 7.5. *In practical scenarios, the update step γ may be selected as a constant value together with a suitably defined stopping criterion evaluating the norm of the gradient at each step, trading asymptotic convergence for implementation simplicity. Consequently, if the magnitude of the update step is large, it could induce oscillations around the solution which could be undesirable or even lead to an auxiliary input that violates Assumption 7.4. Furthermore, the feasibility of Algorithm 1 may also depend on the value of the parameter α. Recalling* (7.54b)*, a large value could make the backward integration numerically unfeasible.* ▲

Remark 7.6. *Recall that, as noted in [45], another parameter of interest concerning the numerical implementations of Algorithm 1 is the sampling interval with respect to which the time interval* $[0, t_f]$ *is discretized. The magnitude of the sampling interval must usually be small to allow for a feasible approximation of the continuous-time* *dynamics involved in Problem 7.1 and has direct repercussions on the computational complexity of the integration steps.* ▲

The choice of the descent step in Algorithm 1 is a crucial design task. Many heuristic techniques have emerged in the literature to improve the convergence of this method, in particular thanks to its extensive diffusion in finite-dimensional optimization fields such as, *e.g.*, machine learning for the training of neural networks [46]. Unlike the finite-dimensional setting, the descent direction provided by the gradient method in the this context is infinite-dimensional. Nonetheless, the choice of the proper descent step constitutes a one-dimensional task in both settings. This makes the use of heuristic methods still viable in the infinite-dimensional setting.

The ADAM scheme yields the update rule given in Algorithm 2. With a temporary abuse of notation, the variables $s(\cdot)$ and $r(\cdot)$ store at each iteration, respectively, the values of $H_v(\cdot)$ and $H_v(\cdot) \odot H_v(\cdot)$ of the current and of the previous iterations for a given value of their argument. Note also that the operation $(\cdot)^{1/2}$ on $\hat{r}(\cdot)$, as well as the operation $(\cdot)^{-1}$ on $((\hat{r}(\cdot))^{1/2} + \delta)$, reported on line 16 of Algorithm 2, are both performed element-wise. The values ρ_1 for s, and ρ_2 for r, constitute a measure of the importance of the past history, which is then used for the proper evaluation of the step size. In particular, an appropriate choice of these parameters allows discarding all the information related to less recent iterations, thus updating the step size by relying only on the most recent information. It is worth mentioning that this scheme adjusts the step size dynamically during the iterations, differently from the scheme provided by Algorithm 1.

The proposed approach has been validated by means of a numerical example. Con-

Algorithm 2 Steady-state ADAM algorithm [44].

Input: $\beta, \gamma, T, t_f, v_0(t), \chi_n(t), \rho_1, \rho_2, \delta$
Output: $\tilde{v}(t)$
let $t \in [0, t_f]$
$\tilde{v}^{(0)}(t) \leftarrow v_0(t)$
$s^{(0)}(t) \leftarrow 0$
$r^{(0)}(t) \leftarrow 0$
for $k = 1, \dots$ **do**
  $\chi_a^{(k)}(t) \leftarrow$ solution of (7.48b) with $v(t) = \tilde{v}^{(k-1)}(t)$
  $(\lambda_n^{(k)}(t), \lambda_a^{(k)}(t)) \leftarrow$ backward integration of (7.54) with
    $(\chi_a(t), v(t)) = (\chi_a^{(k)}(t), \tilde{v}^{(k-1)}(t))$
  if $\delta(H_v(\chi_a^{(k)}(t), \lambda_a^{(k)}(t), \tilde{v}^{(k-1)}(t))) < \beta$ **then return** $\tilde{v}(t) \leftarrow \tilde{v}^{(k-1)}(t)$
  else
    let $\tau \in [0, T]$, $\bar{\tau} = t_f - T + \tau$
    $H_v(\bar{\tau}) \leftarrow H_v(\chi_a^{(k)}(\bar{\tau}), \lambda_a^{(k)}(\bar{\tau}), \tilde{v}^{(k-1)}(\bar{\tau}))$
    $s^{(k)}(\bar{\tau}) \leftarrow \rho_1 s^{(k-1)}(\bar{\tau}) + (1-\rho_1) H_v(\bar{\tau})$
    $r^{(k)}(\bar{\tau}) \leftarrow \rho_2 r^{(k-1)}(\bar{\tau}) + (1-\rho_2) H_v(\bar{\tau}) \odot H_v(\bar{\tau})$
    $\hat{s}(\bar{\tau}) \leftarrow s(\bar{\tau})/(1-\rho_1^k)$
    $\hat{r}(\bar{\tau}) \leftarrow r(\bar{\tau})/(1-\rho_2^k)$
    $\Delta u^{(k)}(\bar{\tau}) \leftarrow -\gamma\, \hat{s}(\bar{\tau}) \odot ((\hat{r}(\bar{\tau}))^{1/2} + \delta)^{-1}$
    $u^{(k)}(\bar{\tau}) \leftarrow \tilde{v}^{(k-1)}(\bar{\tau}) + \Delta u^{(k)}(\bar{\tau})$
    $u_p^{(k)}(\tau) \leftarrow u^{(k)}(\bar{\tau})$
    $\tilde{v}^{(k)}(t) \leftarrow \hat{u}_p^{(k)}(t)$ ▷ Periodic extension.
  end if
end for

sider the following continuous-time exosystem of the form given in eq. (7.2):

$$\dot{w}(t) = \begin{bmatrix} 0 & 1 & 0 \\ -\omega^2 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} w(t), \qquad t \in \mathbb{R}_{\geq 0}, \tag{7.57a}$$

$$r(t) = \begin{bmatrix} w_1(t) + w_3(t) \\ w_2(t) \end{bmatrix}, \qquad t \in \mathbb{R}_{\geq 0}, \tag{7.57b}$$

with $\omega = 2\pi f$, $f = 0.2$, and $w_0 = \begin{pmatrix} 0 & \omega & 2 \end{pmatrix}'$. Then, consider the following continuous-time system of the form (7.1)

$$\dot{\chi}(t) = \begin{bmatrix} 0 \\ \chi_1(t) + \chi_2(t) \\ \chi_1(t) - \chi_2(t) \end{bmatrix} + \begin{bmatrix} e^{\chi_2(t)} & 1 \\ e^{\chi_2(t)} & 1 \\ 0 & 0 \end{bmatrix} (h_c(\chi(t), r(t)) + v(t)), \qquad t \in \mathbb{R}_{\geq 0}, \tag{7.58a}$$

$$\eta_p(t) = h_c(\chi(t), r(t)) + v(t), \qquad t \in \mathbb{R}_{\geq 0}, \tag{7.58b}$$

$$\eta_o(t) = (1 + \chi_3(t))^3, \qquad t \in \mathbb{R}_{\geq 0}, \tag{7.58c}$$

where $h_c : \mathbb{R}^3 \times \mathbb{R}^3 \to \mathbb{R}^2$ is a given feedback control law, and $v : \mathbb{R} \to \mathbb{R}^2$ is the auxiliary input, to be determined as an extremal solution to Problem 7.4. The control objective in this example is that of tracking, with the regulated output $\eta_o(t)$, the exogenous input $r_1(t) = w_1(t) + w_3(t) = \sin(\omega t) + 2$. To highlight the optimization action operated by the auxiliary input v, in a similar way with respect to the previous linear case, the control law $h_c(\chi(t), r(t))$ has been designed as a feedback-linearizing input operating only on the first input, setting the second to zero since an inspection of (7.58) reveals that the system has relative degree $r = 3$ from the first component of the input, affecting the system via the input map $g_1(\chi(t)) = \begin{pmatrix} e^{\chi_2(t)} & e^{\chi_2(t)} & 0 \end{pmatrix}'$, to the output η_o. For a description of the synthesis procedure see [39, Chap. 4].

Algorithm 1 has been tested with parameters $(\gamma,\, t_f) = (10^{-4}, 15)$, $v_0 \equiv 0$, $\chi(0)$ initialized on the manifold described by $\chi(t) = \pi(w(t))$, $t \in \mathbb{R}_{\geq 0}$ consistently with w_0, which has been numerically determined, and disregarding the stopping criterion setting a maximum number of iterations equal to 10^4. The goal of this test is twofold: first, check the performance of the algorithm, and secondly, tune the parameter α. Figure 7.7 shows the steady-state tracking errors achieved with different weights for the invisibility term after the algorithm terminated all the iterations: the problem remains computationally feasible and it is possible to set even a large invisibility weight, *i.e.*,

$\alpha = 10^5$, achieving a low output error.

A comparison has been made, retaining all the parameters, between the rates of convergence of Algorithm 1 and of Algorithm 2, selecting for the latter the parameters ρ_1, ρ_2, and δ as $\rho_1 = 0.9$, $\rho_2 = 0.999$ and $\delta = 10^{-8}$. Figure 7.8 shows how Algorithm 2 is faster than Algorithm 1 in this example, converging in around half the maximum number of iterations allowed and well before the convergence of Algorithm 1.

Figure 7.9 shows the nominal system input given by the feedback linearization process and Figure 7.10 depicts the auxiliary input determined by Algorithm 2. Finally, Figure 7.11 illustrates the performance output subject to the allocation action. It is worth pointing out that, even if the feedback-linearizing law uses only one input, the auxiliary one is distributed among both. Thus, the allocated system input is remodulated among both inputs and, in accordance with the definition of the performance output in this example, η_p exhibits a reduced amplitude leading to a smaller norm, all of which while only a small alteration in the regulated output η_o is observed.

7.6.2 Dynamic allocation with full output invisibility

The discussion of Problem 7.1 in Section 7.6.1 has highlighted a common aspect of all dynamic control allocation problems: the invisibility requirement, formulated either as a problem item or as an optimization constraint, is the hardest one to manage. The steady-state formulation of Problem 7.1 allows the designer to exploit the steady-state behavior of the system, once it is determined, to design the dynamics of the allocator. This is not possible when dealing with Problem 7.2, which requires to account for the full dynamics of (7.1)-(7.2).

The intuition at the heart of the design of dynamic allocators achieving full output

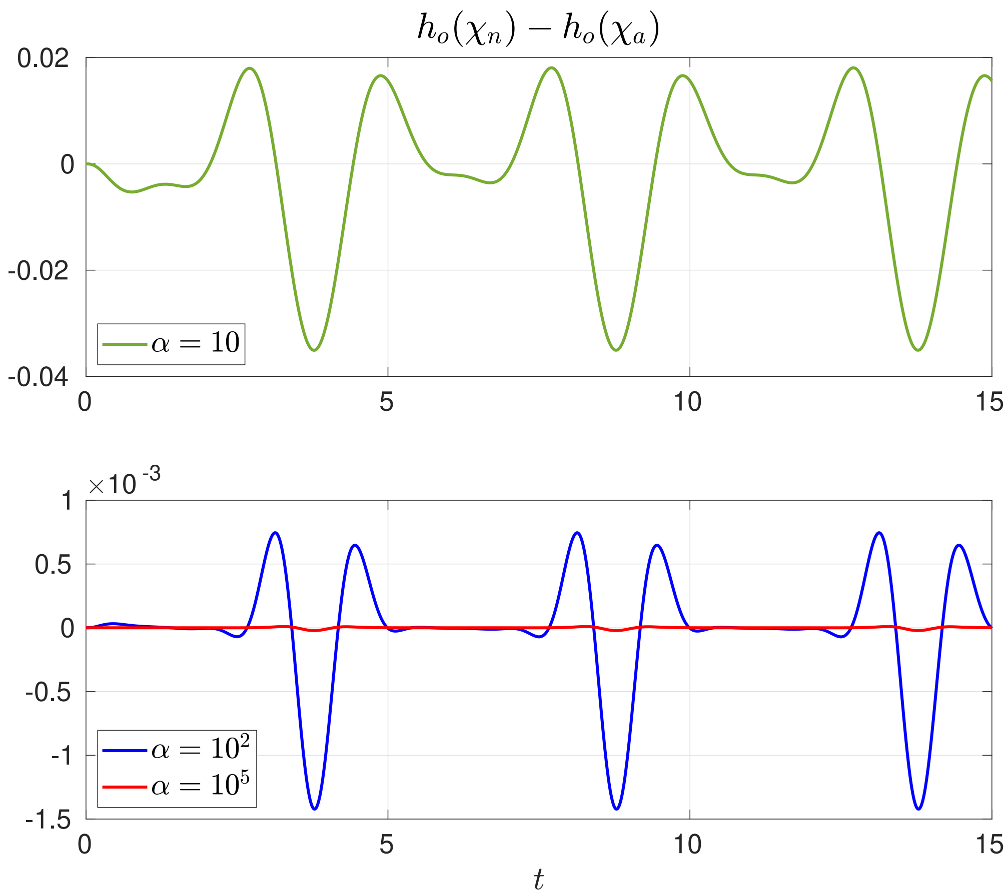

Figure 7.7: Difference between the regulated output with and without input allocation, for different values of the invisibility weight α, during the final run of Algorithm 1.

invisibility is illustrated in [30]. In essence, given items (I) and (O) of Problem 7.2, the allocator dynamics have to be split into multiple subsystems, each of which deals with one of the two requirements. While [30] deals with a problem similar to Problem 7.2 but in which a steady-state cost *function* is considered, its results provide a way to develop an approach suitable for Problem 7.2. An allocator solving Problem 7.2 is composed by, at least, the interconnection of the following two subsystems:

- an *annihilator*, responsible for the full output invisibility requirement and acting as a filter that masks the effect of the allocation action on the regulated output;
- an *optimizer*, responsible for the optimality requirement and acting as a dy-

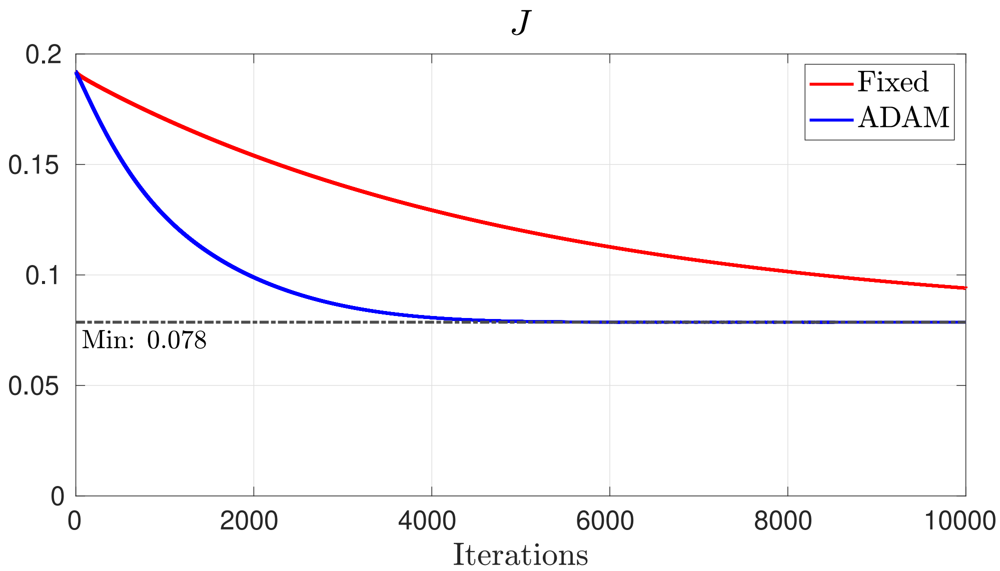

Figure 7.8: Cost during all the iterations performed with Algorithm 1 (red) versus those with Algorithm 2 (blue).

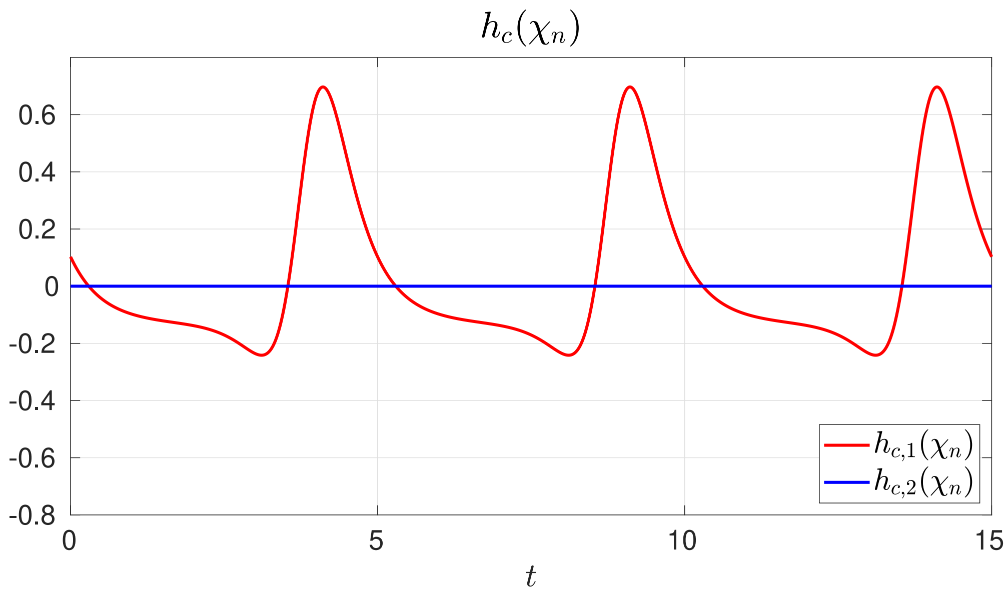

Figure 7.9: Nominal feedback-linearizing system input.

namic controller that computes the allocation action to bring the cost functional (7.5) to its minimal value.

In the case of [30], *i.e.*, when the cost is evaluated only at steady-state and (7.5) reduces to a function, the two subsystems mentioned above are sufficient to achieve the

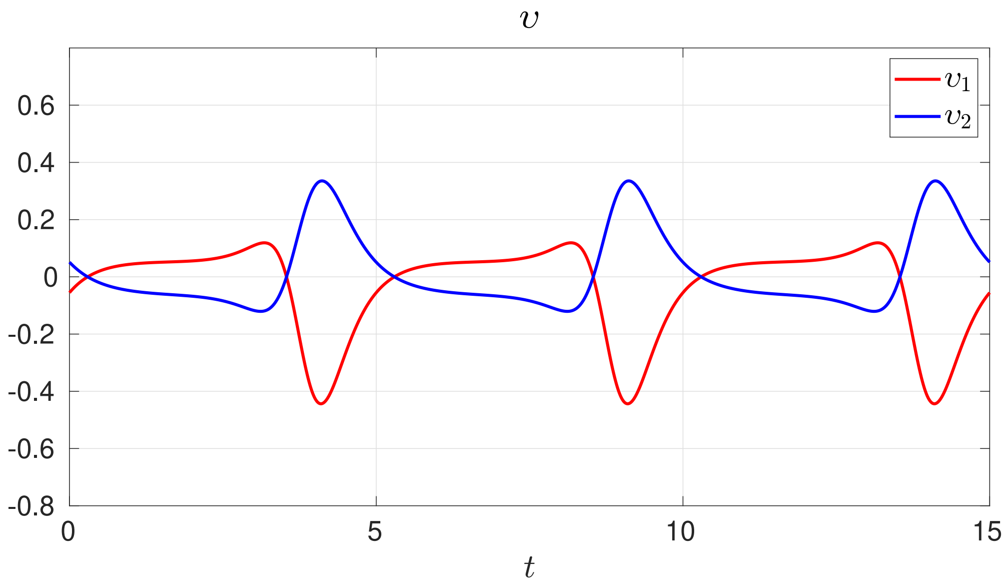

Figure 7.10: Auxiliary input determined by Algorithm 2.

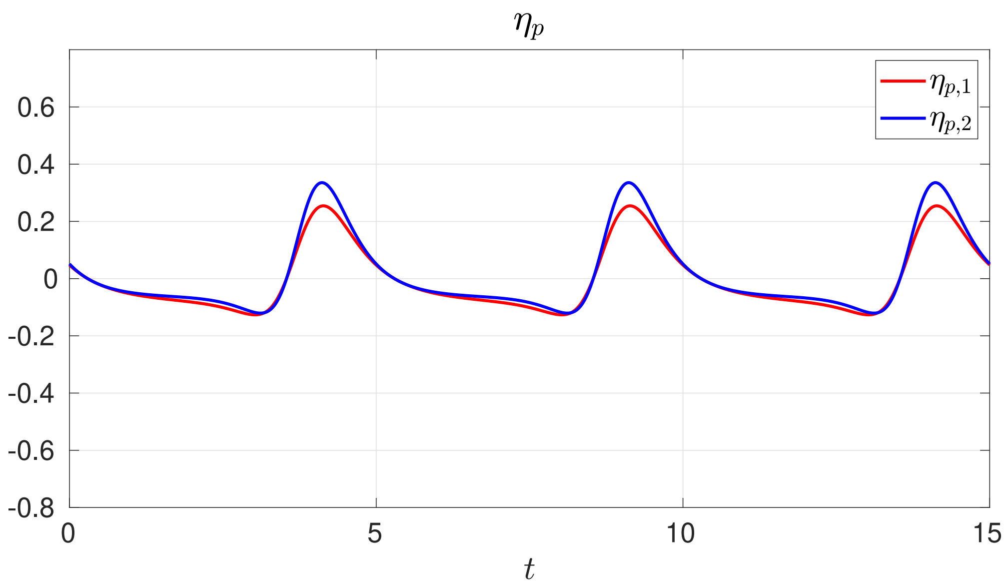

Figure 7.11: Performance output subject to the input allocation determined by Algorithm 2.

desired control objectives. Instead, in the more general setting defined by Problem 7.2, an additional component is necessary to ensure that the steady-state periodicity requirement holds: a *steady-state generator*, which contains an internal model of the exosystem (7.2) and is in charge of producing functions of time within a prescribed set. The interconnection of these three basic components is depicted in Figure 7.12.

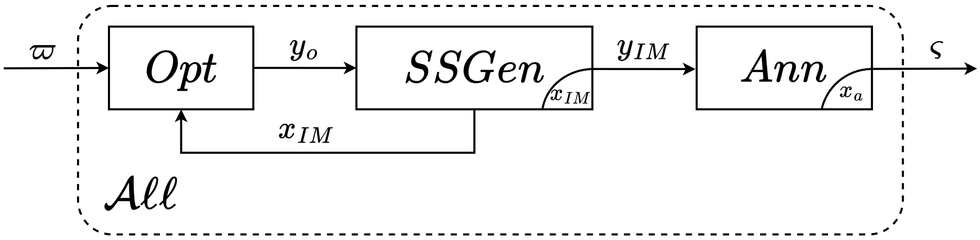

Figure 7.12: Block diagram of a dynamic allocator achieving full output invisibility as requested by Problem 7.2: the optimizer, steady-state generator, and annihilator subsystems are represented by the blocks Opt, *SSGen*, and Ann, respectively.

As highlighted in [30, Remark 4], this construction allows the designer to overcome cumbersome limitations encountered in earlier works that attempt to achieve full output invisibility but fail to do so without relying on quite non-trivial properties of Σ; an example of this for linear systems is found in [24] with the *strong input redundancy* property. Another important aspect of the allocator proposed in [30] is that the design of its three components is carried out in sequence, as summarized in the following steps.

1. The annihilator is designed first, exploiting the dynamical model of the system Σ and its properties. The idea is to design a system whose input-output behavior is *orthogonal* to that of the system Σ, with respect to the regulated output η_o. The results in [30] provide two different ways to design such a component: a polynomial method to compute orthogonal transfer matrices, and a geometric approach based on invariant subspaces.

2. The optimizer is designed next, using a static description of the steady-state behavior of the annihilator that is now available as, *e.g.*, a static gain (see again [42, 43]). The design of this component is aimed at minimizing the steady-state cost in (7.5), so it mainly depends on the formulation of the cost itself.

3. The steady-state generator is designed last, integrating an internal model of the

exosystem E.

This work focuses on the design of an allocator solving Problem 7.2 in the case of a linear model of the form of (7.1)-(7.2) and in two different situations: when the model is fully known and when it is affected by parametric uncertainties. In particular, the design of the annihilator is carried out applying a polynomial approach that is derived from the one discussed in [30] but that is then adapted to the uncertain case. Since the two methods share many similarities, especially in the nominal case, an overview of the original method presented in [30] is omitted; the interested reader is referred to the original work for more insights on the problem formulation thereby considered, as well as on the geometric annihilator design approach.

Consider now a continuous-time, linear time-invariant model given by

$$\dot{\chi}(t) = A(\mu)\chi(t) + B(\mu)v(t) + P(\mu)w(t), \qquad t \in \mathbb{R}_{\geq 0}, \tag{7.59a}$$

$$\eta_o(t) = C_1(\mu)\chi(t) + D_{11}(\mu)v(t) + D_{12}(\mu)w(t), \qquad t \in \mathbb{R}_{\geq 0}, \tag{7.59b}$$

$$\eta_p(t) = C_2(\mu)\chi(t) + D_{21}(\mu)v(t) + D_{22}(\mu)w(t), \qquad t \in \mathbb{R}_{\geq 0}, \tag{7.59c}$$

where $\chi(t) \in \mathbb{R}^n$ is the state vector, $v(t) \in \mathbb{R}^m$ is the auxiliary control input, $w(t) \in \mathbb{R}^{n_e}$ is the reference input, and $\eta_o(t) \in \mathbb{R}^{q_o}$ and $\eta_p(t) \in \mathbb{R}^{q_p}$ are the regulated and the performance outputs, respectively. The reference input is again taken as the full state of the exosystem (7.2), *i.e.*, $r = w$ and $n_r = n_e$. The exosystem E is described by the continuous-time, linear time-invariant model

$$\dot{w}(t) = Sw(t), \quad t \in \mathbb{R}_{\geq 0}, \tag{7.60}$$

with initial condition $w(0) = w_0$. The interconnection is depicted in Figure 7.1.

It is assumed that the matrices in (7.59) depend continuously on an unmeasured vector μ belonging to a set $\mathscr{M}$ that contains the origin: $\mu = 0$ identifies the nominal case. Although in similar cases studied in the literature (see, *e.g.*, [27]), the set $\mathscr{M}$ is assumed to be fully described and compact, these properties are not necessary for the results that follow. To simplify the notation in the following, the dependence of the matrices in (7.59) on μ is omitted unless strictly necessary. The properties of the uncertain model are characterized by the following assumption.

Assumption 7.8. (7.59) *is fully known for* $\mu = 0$ *and the matrix* A *in* (7.59a) *is Hurwitz for any* $\mu \in \mathscr{M}$. *The matrix* $S \in \mathbb{R}^{n_e \times n_e}$ *in* (7.60) *is such that* $\mathrm{spec}(S) \subset \mathbb{C}_0$*; moreover, its eigenvalues are all distinct and possess* commensurate *angular frequencies.* ∘

As an immediate consequence of Assumption 7.8 and of the "blanket" Assumptions 7.1 and 7.2, the exosystem (7.60) is Poisson stable (see Section 7.6.1) and its evolution is periodic of period T defined by the least common multiple of the individual periods.

For $s \in \mathbb{C}$,

$$W_{\eta_o v}(s) = C_1(sI - A)^{-1}B + D_{11}, \tag{7.61a}$$

$$W_{\eta_p v}(s) = C_2(sI - A)^{-1}B + D_{21}, \tag{7.61b}$$

denote the transfer matrices of (7.59) from v to η_o and from v to η_p, respectively. Recalling Assumptions 7.3 and 7.4, it is necessary to further characterize the over-actuation property in the uncertain case introduced by (7.59) to ensure that input allocation remains possible.

Assumption 7.9. *For any* $\mu \in \mathscr{M}$, *the system* (7.59a), (7.59b) *is such that* $m > q_o$, $\mathrm{rank}(B) = m$, *and* $\mathrm{rank}(W_{\eta_o v}(s)) = q_o$. ∘

Finally, recall the definition of the cost functional (7.5) as well as the properties of the instantaneous cost function ℓ. To provide a concise description of the control objective in this scenario,

$$\eta_0(t, t_0, \chi_0, w(\tau), v(\tau)), \quad \tau \in [t_0, t], \tag{7.62}$$

is the output response for output η_o at time $t \geq t_0$ of the plant Σ initialized at $\chi(t_0) = \chi_0$ and driven by the inputs $w(\cdot)$ and $v(\cdot)$. The control problem of interest in this context is formulated as follows.

Problem 7.5. *Consider the systems* (7.59)-(7.60) *and the steady-state cost functional* (7.5). *Suppose that Assumptions 7.4, 7.8 and 7.9 hold. Find, if possible, a* dynamic control allocator *with input ϖ and output ς, to be interconnected to Σ and E according to*

$$\varpi = w, \tag{7.63a}$$

$$\varsigma = v, \tag{7.63b}$$

such that the following requirements are satisfied.

(i) Fixed-time robust output invisibility*: there exists a finite time* $t_0 \geq 0$ *with the property that, for any $\mu \in \mathcal{M}$, the regulated output η_o satisfies*

$$\eta_0(t, t_0, \chi_0, w(\tau), 0) = \eta_0(t, t_0, \chi_0, w(\tau), v(\tau)), \quad t \geq t_0, \tau \in [t_0, t], (\chi_0, w_0) \in \mathbb{R}^n \times \mathscr{W}^\circ. \tag{7.64}$$

(ii) Steady-state periodicity*: for any $w_0 \in \mathscr{W}^\circ$, the internal state of the control allocator is bounded for any time and its unique steady-state response is periodic of period T.*

(iii) Optimality*: the Lagrange cost functional* (7.5) *is minimized with respect to the class of inputs in a* prescribed *finite-dimensional function space.*

The construction of a solution to Problem 7.5 for any $\mu \in \mathscr{M}$ begins by identifying the data whose knowledge is necessary for the design of the allocator. This knowledge is obtained in the uncertain case by means of algorithms that are described in the following and are based on input-output measurements. Let the spectrum of the matrix S in (7.60) be

$$\mathrm{spec}(S) = \{0, j\omega_1, -j\omega_1, ..., j\omega_{\bar{n}_e}, -j\omega_{\bar{n}_e}\}, \quad \bar{n}_e = \frac{n_e - 1}{2}. \tag{7.65}$$

Define the *degree of redundancy* of (7.59) with respect to the output η_o as

$$m_a = m - q_o. \tag{7.66}$$

The design of the allocator requires the following information.

Definition 7.1 (Data for the dynamic control allocator (7.75))**.** *The data necessary for the design of the dynamic control allocator* (7.75) *is divided into three items.*

K1 *The* steady-state map $\bar{M}_{\eta_p,w}$ *from* w *to* η_p, i.e.*, a matrix such that, denoting by* $\tilde{\eta}_p^w(\cdot)$ *the part of the steady-state response in* η_p *due to* w*,*

$$\tilde{\eta}_p^w(t) = \bar{M}_{\eta_p,w} w(t), \quad t \in \mathbb{R}_{\geq 0}. \tag{7.67}$$

K2 *The* moments *of the transfer matrix* $W_{\eta_p v}(s)$ *of* (7.59) *at* $s \in \{0, j\omega_1,, ..., j\omega_{\bar{n}_e}\} \subset \mathrm{spec}(S)$*, namely*

$$W_{\eta_p v}(0), W_{\eta_p v}(j\omega_1), \dots, W_{\eta_p v}(j\omega_{\bar{n}_e}). \tag{7.68}$$

K3 *A* polynomial annihilator *for the transfer matrix* $W_{\eta_o v}(s)$ *of* (7.59)*, namely* $W_a(s) \in \mathbb{C}^{m\times m_a}$*, with poles with negative real part, such that*

$$W_{\eta_o v}(s)W_a(s) = 0, \quad s \in \mathbb{C} \setminus (\mathscr{P}(W_{\eta_o v}(s)) \cup \mathscr{P}(W_a(s))). \tag{7.69}$$

∘

If the data **K1**, **K2**, and **K3** are available, the proposed dynamic allocator can be constructed as follows. Let (A_a, B_a, C_a) be a minimal realization of $W_a(s)$ in (7.69). Define

$$\Omega = \text{diag}\left(0, \begin{bmatrix} 0 & \omega_1 \\ -\omega_1 & 0 \end{bmatrix}, \dots, \begin{bmatrix} 0 & \omega_{\bar{n}_e} \\ -\omega_{\bar{n}_e} & 0 \end{bmatrix}\right), \tag{7.70a}$$

$$\Lambda = \left[\, 1 \,\vert\, 1 \;\; 0 \,\vert \cdots \vert\, 1 \;\; 0 \,\right], \tag{7.70b}$$

yielding the matrices (A_{IM}, C_{IM}):

$$A_{IM} = \Omega \otimes I_{m_a}, \tag{7.71a}$$

$$C_{IM} = \Lambda \otimes I_{m_a}. \tag{7.71b}$$

Furthermore, using the data **K2**, define

$$W_{\eta_p \varpi}(j\omega) = W_{\eta_p v}(j\omega)W_a(j\omega), \quad \omega \in \{0, \omega_1, \dots, \omega_{\bar{n}_e}\}, \tag{7.72}$$

and build the matrix $\bar{N}_\mu$ as

$$\bar{N}_\mu = \left[W_{\eta_p\varpi}(0) \,\middle|\, \mathrm{Re}(W_{\eta_p\varpi}(j\omega_1)) \;\; \mathrm{Im}(W_{\eta_p\varpi}(j\omega_1)) \,\middle|\, \cdots \,\middle|\, \mathrm{Re}(W_{\eta_p\varpi}(j\omega_{\bar{n}_e})) \;\; \mathrm{Im}(W_{\eta_p\varpi}(j\omega_{\bar{n}_e})) \right]. \tag{7.73}$$

Define the functions $\xi(t)$ and $L(w(t), x_{IM}(t))$ as

$$\xi(t) = \bar{M}_{\eta_p,w} e^{St} w(t) + \bar{N}_\mu e^{A_{IM}t} x_{IM}(t), \qquad t \in \mathbb{R}_{\geq 0}, \tag{7.74a}$$

$$L(w(t), x_{IM}(t)) = \left(\int_0^T \nabla \ell(\xi(\sigma))' \bar{N}_\mu e^{A_{IM}\sigma} d\sigma \right)', \qquad t \in \mathbb{R}_{\geq 0}. \tag{7.74b}$$

The proposed dynamic allocator is given by

$$\dot{x}_{IM}(t) = A_{IM} x_{IM}(t) - \gamma L(w(t), x_{IM}(t)), \qquad t \in \mathbb{R}_{\geq 0}, \tag{7.75a}$$

$$\dot{x}_a(t) = A_a x_a(t) + B_a C_{IM} x_{IM}(t), \qquad t \in \mathbb{R}_{\geq 0}, \tag{7.75b}$$

$$\varsigma(t) = C_a x_a(t), \qquad t \in \mathbb{R}_{\geq 0}. \tag{7.75c}$$

Remark 7.7. *The equations* (7.74)-(7.75) *allow the designer to directly implement the proposed allocation scheme (more insights on how to do this in software routines as well as on devices such as,* e.g.*, digital microcontrollers, are given in the following). Nonetheless, a deeper understanding of* (7.75) *is necessary to clarify its construction with respect to the architecture introduced in [30], anticipating the following results. The optimizer unit is given by* (7.74) *and it embeds, in essence, a gradient descent scheme as shown in the following. The state vector $x_{IM}(t)$ is the state of the steady-state generator, whose dynamics are defined by* (7.70)-(7.71). *The state vector $x_a(t)$ is the state of the polynomial annihilator given by $W_a(s)$ and the matrices (A_a, B_a, C_a) are a minimal realization of it.* ▲

Remark 7.8. *The definition of A_{IM} given in* (7.71a)*, together with its eigenvalues selected by* (7.70a)*, define the* prescribed *function space mentioned in item (iii) of Problem 7.5: it consists of the functions of time generated as combinations of the modes of* (7.60)*,* i.e.*, any function $v(t)$ such that there exist constants c_0, c_i, ϕ_i, $i = 1, \ldots, \bar{n}_e$, with the property that $v(t) = c_0 + \sum_{i=1}^{\bar{n}_e} c_i \sin(\omega_i t + \phi_i)$.* ▲

Remark 7.9. *In [29, Problem 1], a formulation similar to Problem 7.5 is provided, although concerning the only case of $\mu = 0$. The results shown therein lead to the development of a dynamic control allocator that is implemented as a hybrid dynamical system, which has also been compared against the solution of the optimal control approach presented in Section 7.6.1. With respect to that work, the solution proposed in* (7.75) *has two main advantages. First, it is a continuous-time system with no hybrid dynamics, which also simplifies its implementation. Secondly, the solution presented in [29] is based on a gradient descent approach, as is the one hereby proposed (see* (7.75a)*), and both these solutions depend on a gradient descent step paramter $\gamma \in \mathbb{R}_{>0}$. What differentiates the proposed approach from the state of the art is that, while in [29] the step parameter γ has to be tuned carefully to ensure convergence due to the hybrid dynamics, here it can be set* arbitrarily large*, thus leading to an arbitrarily fast allocation action.* ▲

The equations (7.75) allow the designer to directly implement the proposed control allocator. The properties of this solution are clarified in the following. In the case of $\mu = 0$, the gathering of the data **K1**, **K2**, and **K3** is straightforward.

Remark 7.10 (**K1**, **K2**, **K3** for $\mu = 0$)**.** *If $\mu = 0$,* (7.59) *is fully known by Assumption 7.8 and the data **K1**, **K2**, and **K3** is computed as follows.*

K1 *Solve the following:*

$$\Pi_\Sigma S = A\Pi_\Sigma + P, \tag{7.76a}$$

$$\bar{M}_{\eta_p,w} = C_2\Pi_\Sigma + D_{22}. \tag{7.76b}$$

K2 $\{W_{\eta_p v}(0), W_{\eta_p v}(j\omega_1), \ldots, W_{\eta_p v}(j\omega_{\bar{n}_e})\}$ *are given by*

$$W_{\eta_p v}(s), \quad s \in \{0, j\omega_1, \ldots, j\omega_{\bar{n}_e}\}. \tag{7.77}$$

K3 *Find $N_a(s)$ whose columns are a minimal polynomial basis for the null space of $W_{\eta_o v}(s)$. Choose a Hurwitz monic polynomial $d_a(s)$ of degree larger than each element of $N_a(s)$. Define*

$$W_a(s) = d_a^{-1}(s)N_a(s), \tag{7.78}$$

and choose $(A_a, B_a, C_a, 0)$ as any minimal realization of $W_a(s)$. ▲

The following statement characterizes the properties of (7.75) in the case of $\mu = 0$.

Lemma 7.4. *Consider the system* (7.59) *with $\mu = 0$. Then, the dynamic control allocator* (7.75) *interconnected to* (7.59) *as in* (7.63) *solves Problem 7.5 for any $\gamma \in \mathbb{R}_{\geq 0}$ with $t_0 = 0$.* ∘

Proof. Item (i) of Problem 7.5 with $\mu = 0$ follows immediately by knowledge of **K3** computed as in Remark 7.10; this guarantees full output invisibility of the allocator unit as requested. Therefore, the optimality requirement remains to be addressed. To this end, it must be shown that the behavior of the proposed allocator (7.75) leads to the minimization of the cost functional (7.5). This is achieved by showing that (7.75a), (7.74b) define a gradient descent algorithm for the cost functional (7.5). Define the *moving-horizon cost* over a period as

$$J_t = \int_t^{t+T} \ell(\tilde{\eta}_p(\sigma))\, d\sigma, \quad t \in \mathbb{R}_{\geq 0}, \tag{7.79}$$

and consider (7.75a) with an input $u_{IM}(t)$ to be designed, *i.e.*,

$$\dot{x}_{IM}(t) = A_{IM} x_{IM}(t) + u_{IM}(t), \quad t \in \mathbb{R}_{\geq 0}. \tag{7.80}$$

To evaluate the effect of a change in the cost J due to a change in x_{IM}, consider the following selections of u_{IM}.

$$u_{IM}^{\delta}(\sigma) = \begin{cases} \delta, & \text{if } \sigma \in [t, t+\Delta],\, \Delta > 0,\, t \in \mathbb{R}_{\geq 0}, \\ 0, & \text{otherwise.} \end{cases} \tag{7.81a}$$

$$u_{IM}^{\circ}(\sigma) = 0, \quad \forall \sigma \geq t,\, t \in \mathbb{R}_{\geq 0}. \tag{7.81b}$$

Denote the respective evolutions of $\tilde{\eta}_p$, x_{IM}, and J_t by the superscripts δ or $\circ$, *e.g.*, $\tilde{\eta}_p^{\delta}$ and $\tilde{\eta}_p^{\circ}$. Integrating (7.80), (7.81) from the common initial condition $x_{IM}^{\circ}(t)$ yields

$$x_{IM}^{\circ}(t+\Delta) = e^{A_{IM}\Delta} x_{IM}^{\circ}(t), \tag{7.82a}$$

$$\begin{aligned} x_{IM}^{\delta}(t+\Delta) &= e^{A_{IM}\Delta} x_{IM}^{\circ}(t) + \int_0^{\Delta} e^{A_{IM}\tau} u_{IM}(t+\Delta-\tau)\, d\tau \\ &= x_{IM}^{\circ}(t+\Delta) + \int_0^{\Delta} e^{A_{IM}\tau}\, d\tau \delta = x_{IM}^{\circ}(t+\Delta) + \Delta \bar{B}\delta. \end{aligned} \tag{7.82b}$$

The matrix $\bar{B}$ is such that

$$\|\bar{B}\| \leq 1, \qquad \lim_{\Delta \to 0^+} \bar{B} = I, \tag{7.83}$$

which follow from the fact that $\|e^{A_{IM}\tau}\| = 1$ (since $(e^{A_{IM}\tau})' e^{A_{IM}\tau} = I$) and from integration of the series expansion of $e^{A_{IM}\tau}$, respectively. Since u°_{IM} and u^{δ}_{IM} are zero after $t+\Delta$, it holds that

$$x^{\delta}_{IM}(\tau) - x^{\circ}_{IM}(\tau) = e^{A_{IM}(\tau-(t+\Delta))}\Delta\bar{B}\delta, \quad \tau \geq t+\Delta, \tag{7.84}$$

and the corresponding steady-state responses of η_p are

$$\tilde{\eta}^{\circ}_p(\tau) = \bar{M}_{\eta_p,w}w(\tau) + \bar{N}_\mu x^{\circ}_{IM}(\tau), \tag{7.85a}$$

$$\tilde{\eta}^{\delta}_p(\tau) = \bar{M}_{\eta_p,w}w(\tau) + \bar{N}_\mu x^{\delta}_{IM}(\tau), \tag{7.85b}$$

$$\hat{\tilde{\eta}}_p(\tau) = \tilde{\eta}^{\delta}_p(\tau) - \tilde{\eta}^{\circ}_p(\tau) = \bar{N}_\mu e^{A_{IM}(\tau-(t+\Delta))}\Delta\bar{B}\delta. \tag{7.85c}$$

If Δ is sufficiently small, the difference between the two cost integrand functions along the two solutions satisfy

$$\begin{aligned}\ell(\tilde{\eta}^{\delta}_p(\tau)) - \ell(\tilde{\eta}^{\circ}_p(\tau)) &= \nabla\ell|_{\tilde{\eta}^{\circ}_p(\tau)}\,\hat{\tilde{\eta}}_p(\tau) + o(\|\hat{\tilde{\eta}}_p(\tau)\|) \\ &= \nabla\ell|_{\tilde{\eta}^{\circ}_p(\tau)}\,\bar{N}_\mu e^{A_{IM}(\tau-(t+\Delta))}\Delta\bar{B}\delta + o(\|\Delta\|),\end{aligned} \tag{7.86}$$

where the equality $o(\|\hat{\tilde{\eta}}_p(\tau)\|) = o(\|\Delta\|)$ follows from (7.85c), (7.83), implying in turn that

$$\|\bar{N}_\mu e^{A_{IM}(\tau-(t+\Delta))}\Delta\bar{B}\delta\| \leq \|\bar{N}_\mu\|\|\Delta\|\|\delta\|, \tag{7.87}$$

provided that $||\delta||$ is bounded by a quantity independent of Δ, which is shown in the following. Periodicity of $\tilde{\eta}^{\circ}_p$ yields

$$J^{\circ}_t = J^{\circ}_{t+\Delta}, \quad t \in \mathbb{R}_{\geq 0}, \tag{7.88}$$

hence, for small Δ:

$$
\begin{aligned}
J^{\delta}_{t+\Delta} - J^{\circ}_{t} &= J^{\delta}_{t+\Delta} - J^{\circ}_{t+\Delta} \\
&= \int_{t+\Delta}^{t+\Delta+T} \left(\ell(\tilde{\eta}^{\delta}_{p}(\tau)) - \ell(\tilde{\eta}^{\circ}_{p}(\tau))\right) d\tau \\
&= \int_{t+\Delta}^{t+\Delta+T} \left(\nabla\ell|_{\tilde{\eta}^{\circ}_{p}(\tau)}\, \hat{\tilde{\eta}}_{p}(\tau) + o(\|\Delta\|)\right) d\tau \\
&= \int_{t+\Delta}^{t+\Delta+T} \nabla\ell|_{\tilde{\eta}^{\circ}_{p}(\tau)}\, \hat{\tilde{\eta}}_{p}(\tau)\, d\tau + o(\|\Delta\|).
\end{aligned}
\tag{7.89}
$$

Hence, recalling that $\lim_{\Delta\to 0^+} \frac{1}{\Delta} o(\|\Delta\|) = 0$ by definition of $o(\cdot)$, and (7.85c), it follows that

$$
\begin{aligned}
\dot{J}_t &= \lim_{\Delta\to 0^+} \frac{J^{\delta}_{t+\Delta} - J^{\circ}_{t}}{\Delta} \\
&= \lim_{\Delta\to 0^+} \frac{1}{\Delta} \int_{t+\Delta}^{t+\Delta+T} \nabla\ell|_{\tilde{\eta}^{\circ}_{p}(\tau)}\, \hat{\tilde{\eta}}_{p}(\tau)\, d\tau \\
&= \int_{t}^{t+T} \nabla\ell|_{\tilde{\eta}^{\circ}_{p}(\tau)}\, \bar{N}_{\mu} e^{A_{IM}(\tau-t)}\, d\tau\delta \\
&= F(t)\delta, \quad t \in \mathbb{R}_{\geq 0}.
\end{aligned}
\tag{7.90}
$$

Selecting $\delta = -\gamma F(t)$ for any $\gamma \in \mathbb{R}_{\geq 0}$ yields

$$
\dot{J}_t = -\gamma \|F(t)\|^2, \quad t \in \mathbb{R}_{\geq 0}. \tag{7.91}
$$

Note that $F(t)$ only depends on the nominal $\tilde{\eta}^{\circ}_{p}$ trajectory and is independent of Δ, as required after (7.87). The change of the variable $\sigma = \tau - t$ in the definition of $F(t)$ shows that $F(t)' = L(w(t), x_{IM}(t))$ with the latter defined in (7.74). The latter equation (7.91), in turn, together with boundedness of $\bar{M}_{\eta_p,w}w(t)$ for any time as in (7.67), implies that x_{IM} in (7.75a) is bounded and that, at steady-state, it is described by the differential equation $\dot{\tilde{x}}_{IM}(t) = A_{IM}\tilde{x}_{IM}(t)$, the solution of which is, by construction, periodic of period T. Finally, the annihilator is asymptotically stable since the matrix A_a in (7.75b) is Hurwitz; hence, the condition on $\dot{J}_t$ implies that items (ii) and (iii) of

Problem 7.5 hold, concluding the proof. □

Remark 7.11. *In practice, the dynamic control allocation scheme* (7.75) *can be implemented by approximating the integral in* (7.74b) *with a finite sum. Given a sufficiently small sampling time* $\tau_s \in \mathbb{R}_{>0}$, $\tau_s = \frac{T}{K}$, $K \in \mathbb{N}$, *define* $\bar{S} = e^{S\tau_s}$ *and* $\bar{A}_{IM} = e^{A_{IM}\tau_s}$. *Then, the gradient of the cost functional can be approximated as*

$$\xi_k = \bar{M}_{\eta_p,w}\bar{S}^k w(k\tau_s) + \bar{N}_\mu \bar{A}_{IM}^k x_{IM}(k\tau_s), \tag{7.92a}$$

$$L(w(t), x_{IM}(t))' \approx \sum_{k=1}^{K} \nabla \ell(\xi_k)' \cdot \bar{N}_\mu \bar{A}_{IM}^k, \quad t \in \mathbb{R}_{\geq 0}, \tag{7.92b}$$

which is a nonlinear function of the measurements of $x_{IM}(t)$ *and* $w(t)$. *The core difficulties in the implementation of* (7.75) *arise in the nonlinear terms of* (7.92) *and fall within the wide spectrum of problems concerning function approximation with digital devices. In this specific case, assuming that the quantities* $w(t)$ *and* $x_{IM}(t)$ *remain constant during sampling intervals, a digital device with a sampling time equal to* τ_s *in* (7.92) *can be used to implement the control allocator* (7.75). ▲

The following result reveals that the structure of the control allocator (7.75) is simplified significantly when the cost funcional (7.5) is quadratic: the correction term (7.74b)-(7.74a) becomes a *static linear feedback* from the state vector in (7.75a) and a *feedforward* from the state of the exosystem (7.60).

Proposition 7.1. *Consider the control allocator* (7.75) *and suppose that* $\ell(\tilde{\eta}_p(t)) = \frac{1}{2}\|\tilde{\eta}_p(t)\|_R^2$, *with* $R = R' > 0$. *Then,* $L(w(t), x_{IM}(t))$ *in* (7.74a)-(7.74b) *becomes*

$$L(w(t), x_{IM}(t)) = H_1 x_{IM}(t) + H_2 w(t), \quad t \in \mathbb{R}_{\geq 0}, \tag{7.93}$$

with

$$H_1 = \int_0^T e^{A'_{IM}\sigma} \bar{N}'_\mu R \bar{N}_\mu e^{A_{IM}\sigma}\, d\sigma, \tag{7.94a}$$

$$H_2 = \int_0^T e^{A'_{IM}\sigma} \bar{N}'_\mu R \bar{M}_{\eta_p,w} e^{S\sigma}\, d\sigma. \tag{7.94b}$$

Hence, for any $\gamma \in \mathbb{R}_{>0}$, (7.75a) *becomes*

$$\dot{x}_{IM}(t) = (A_{IM} - \gamma H_1)x_{IM}(t) - \gamma H_2 w(t), \quad t \in \mathbb{R}_{\geq 0}. \tag{7.95}$$

Moreover, if the pair $(A_{IM}, R^{\frac{1}{2}}\bar{N}_\mu)$ *is observable, then the matrix* $(A_{IM} - \gamma H_1)$ *is Hurwitz for any* $\gamma \in \mathbb{R}_{>0}$. ∘

Proof. In this case, the gradient of the cost function ℓ in (7.5) is linear:

$$\nabla \ell(\tilde{\eta}_p(t)) = R\tilde{\eta}_p(t), \quad t \in \mathbb{R}_{\geq 0}. \tag{7.96}$$

Applying standard properties of integrals, (7.74b) reduces to

$$\begin{aligned} L(w(t), x_{IM}(t)) &= \int_0^T e^{A'_{IM}\sigma} \bar{N}'_\mu R \bar{N}_\mu e^{A_{IM}\sigma}\, d\sigma\, x_{IM} + \int_0^T e^{A'_{IM}\sigma} \bar{N}_\mu R \bar{M}_{\eta_p,w} e^{S\sigma}\, d\sigma\, w(t) \\ &= H_1 x_{IM}(t) + H_2 w(t), \quad t \in \mathbb{R}_{\geq 0}. \end{aligned} \tag{7.97}$$

The last claim of Proposition 7.1 follows from two facts: first, H_1 is the observability Gramian matrix of the pair $(A_{IM}, R^{\frac{1}{2}}\bar{N}_\mu)$, and secondly, the matrix A_{IM} is Hurwitz by construction. □

Remark 7.12. *In practice, the matrices* H_1 *and* H_2 *in* (7.94) *can be computed offline.*

Define

$$A_{12}^1 = \bar{N}_\mu' R \bar{N}_\mu, \tag{7.98a}$$

$$A_{22}^1 = A_{IM}, \tag{7.98b}$$

$$A_{12}^2 = \bar{N}_\mu' R \bar{M}_{\eta_p,w}, \tag{7.98c}$$

$$A_{22}^2 = S, \tag{7.98d}$$

and compute the matrix exponentials

$$H_i = \begin{bmatrix} I & 0 \end{bmatrix} \exp\left(\begin{bmatrix} -A_{IM}' & A_{12}^i \\ 0 & A_{22}^i \end{bmatrix} T \right) \begin{bmatrix} 0 \\ I \end{bmatrix}, \quad i \in \{1,2\}. \tag{7.99}$$

Furthermore, denoting by $x_{IM}^\star(t)$ *the optimal steady-state evolution of* $x_{IM}(t)$, *the fact that the gradient* $L(w(t), x_{IM}^\star(t))$ *is equal to zero for any* $t \in \mathbb{R}_{\geq 0}$ *yields*

$$x_{IM}^\star(t) = -H_1^{-1} H_2 w(t), \quad t \in \mathbb{R}_{\geq 0}. \tag{7.100}$$

The first consequence of (7.100) *is that the correct initialization for the steady-state generator is* $x_{IM}(0) = -H_1^{-1} H_2 w(0)$. *Secondly, one may* bypass *the steady-state generator entirely by redefining the control allocator* (7.75) *as*

$$\dot{x}_a(t) = A_a x_a(t) - B_a C_{IM} H_1^{-1} H_2 w(t), \qquad t \in \mathbb{R}_{\geq 0}, \tag{7.101a}$$

$$y_a(t) = C_a x_a(t), \qquad t \in \mathbb{R}_{\geq 0}. \tag{7.101b}$$

▲

Remark 7.13. *The observability property in the last sentence of Proposition 7.1 is always satisfied if the columns of the numerator of the annihilator* $W_a(s)$, *namely* $N_a(s)$, *are* *chosen as a minimal basis, as it can be inferred from [47, Thm. 6.5.10].* ▲

Dealing with the general case of $\mu \in \mathscr{M}$ requires data-driven algorithms for the estimation of the data in Definition 7.1. To this end, the steady-state behavior of the uncertain system (7.59)-(7.60) must first be estimated. The following results and algorithms provide a solution to this problem based on the concept of *moment matching* for a linear system; the basic definitions and results required in the following are given in Appendix C. Starting from **K1**, the first item to be estimated is the steady-state map $\bar{M}_{\eta_p,w}$. Usually, this implies disconnecting the exosystem (7.59) from the system (7.59); in the current context this procedure is not permitted, since one of the main goals of the allocation approach is to not alter or reconfigure in any way the preexisting control system. Thus, a data-driven algorithm for the online estimation of $\bar{M}_{\eta_p,w}$ with $v = 0$ is proposed. Since it is based on sampled measurements of $\tilde{\eta}_p(t)$ and $w(t)$, the first step consists in defining the characteristics of the sampling procedure. Let $\tau_s \in \mathbb{R}_{>0}$ be the sampling time, characterized by the following statements.

Definition 7.2 (Non-pathological sampling time)**.** *The sampling time* $\tau_s > 0$ *is* non-pathological *if, for any* $\lambda_1, \lambda_2 \in \mathrm{spec}(S) \cup \mathrm{spec}(A)$*, there is no* $h \in \mathbb{Z}$ *such that*

$$\tau_s(\lambda_1 - \lambda_2) = h 2\pi j. \tag{7.102}$$

∘

Assumption 7.10. τ_s *is non-pathological.* ∘

Let $S_d = e^{S\tau_s}$, $A_d = e^{A\tau_s}$. Assumption 7.10 implies that all distinct eigenvalues of S and A are mapped to distinct eigenvalues of S_d and A_d; moreover all structural properties (*e.g.*, observability and reachability) of (7.59)-(7.60) are preserved in their sampled data equivalents. More details about this topic are found in [48, Sec. 3.2].

Algorithm 3 Data-driven pseudo-steady-state output map estimation.

Input: τ_s, $K \in \{a \in \mathbb{N} : a \geq n+1+q_p n_e\}$, $\{w_k\}_{k=0}^{K-1}$, $\{\eta_{p,k}\}_{k=0}^{K+n-1}$.
Output: $\bar{M}_{\eta_p,w}$.

1: Define $T(\eta_p, w)$ as:

$$T(\eta_p, w) = \begin{bmatrix} \eta_p(n\tau_s) & \dots & \eta_p(0) & -w(0)' \otimes I_{q_p} \\ \eta_p((n+1)\tau_s) & \dots & \eta_p(\tau_s) & -w(\tau_s)' \otimes I_{q_p} \\ \vdots & \dots & \vdots & \vdots \\ \eta_p((n+K-1)\tau_s) & \dots & \eta_p((K-1)\tau_s) & -w((K-1)\tau_s)' \otimes I_{q_p} \end{bmatrix}. \tag{7.103}$$

2: Find the smallest $\nu \in \mathbb{N}$, $1 \leq \nu \leq n$, for which there exists $v \in \mathbb{R}^{n+1+q_p n_e}$ satisfying:

$$T(\eta_p, w)\, v = 0, \tag{7.104}$$

with v such that

$$v' = \begin{bmatrix} 0_{1\times(n-\nu)} & 1 & \hat{a}_{\nu-1} & \hat{a}_{\nu-2} & \cdots & \hat{a}_0 & \hat{M}' \end{bmatrix}. \tag{7.105}$$

3: Define $\tilde{M} \in \mathbb{R}^{q_p \times n_e}$ such that $\mathrm{vec}(\tilde{M}) = \hat{M}$.

4: Let $\hat{a}_\nu = 1$ and compute $\bar{M}_{\eta_p,w}$ as:

$$\bar{M}_{\eta_p,w} = \tilde{M}\left(\sum_{h=0}^{\nu} \hat{a}_h S_d^h\right)^{-1}. \tag{7.106}$$

Remark 7.14. *Algorithm 3 holds onto Assumptions 7.8 and 7.10. Moreover, here it is defined for data sampled at times $k = 0, 1, \dots, K-1$ but it can be applied (with obvious changes) to any sequence of K data samples,* i.e.*, for $k = \bar{k}, \dots, \bar{k}+K-1$ for an arbitrary $\bar{k} \in \mathbb{Z}$.* ▲

Proposition 7.2. *Let Assumptions 7.8 and 7.10 hold. Fix $w_0 \in \mathscr{W}^\circ$. Then, the following statements hold.*

i. Algorithm 3 can be applied, yielding a matrix $\bar{M}_{\eta_p,w}$.

ii. $\bar{M}_{\eta_p,w}$ *satisfies* $\tilde{\eta}_p(t) = \bar{M}_{\eta_p,w} w(t)$ *for all* $t \in \mathbb{R}_{\geq 0}$.

iii. If w_0 *excites all natural modes of* (7.60)*, then* $\bar{M}_{\eta_p,w}$ *is equal to the unique solution of* (7.76b)*.* ∘

Note that statement (ii) of Proposition 7.2 highlights that the matrix $\bar{M}_{\eta_p,w}$ computed by Algorithm 3 always ensures a correct prediction of the steady-state evolution *for the trajectory used in the computation*, even if such matrix does not coincide with the unique solution of (7.76b). Further conditions, concerning the quantity of information that the data provides, are required to ensure the conincidence of the two matrices; this is specified in statement (ii) of Proposition 7.2. Since the algorithm is intended to be used for data-driven, online estimation, the trajectory used in the computation (arising from the fixed w_0) is exactly the one of interest for correctly controlling the plant, hence statement (ii) is the one actually useful in practice. The issue at hand is the same as the one in classic adaptive regulation, where the control objective is achieved even without fully identifying the parameters of the plant. Statement (iii) is given to complete the analysis and provide a full understanding of the issues at play.

Proof. To prove item (i) of Proposition 7.2, notice that all steps in Algorithm 3 can be performed, provided that (7.104) is solvable and the matrix inverse in (7.106) exists; both facts are shown in the following. In the following, given a signal $x(t)$, $x^\circ(t)$ denotes its transient response, provided the latter exists. For all $k \in \mathbb{N}$, the following holds:

$$\eta_p(k\tau_s) = \eta_p^\circ(k\tau_s) + \tilde{\eta}_p(k\tau_s) = C_2\chi^\circ(k\tau_s) + \bar{M}_{\eta_p,w} w(k\tau_s). \tag{7.107}$$

Recall [47, App. A.8] that a matrix satisfies the equation defined by its *minimal polynomial* (*i.e.*, the smallest degree polynomial which enjoys this property), and that the minimal polynomial has the same roots, with lower or equal multiplicity, of the *characteristic polynomial*. Define $\hat{a}_\nu = 1$ and let the minimal (monic) polynomial of A_d be:

$$\rho_{A_d}(z) = \sum_{h=0}^{\nu} \hat{a}_h z^h, \quad z \in \mathbb{C}. \tag{7.108}$$

Since $\rho_{A_d}(A_d) = 0$ by definition and

$$w((k+h)\tau_s) = S_d^h w(k\tau_s), \qquad k \in \mathbb{Z}_{\geq 0}, \tag{7.109a}$$

$$\chi^\circ((k+h)\tau_s) = A_d^h \chi^\circ(k\tau_s), \qquad k \in \mathbb{Z}_{\geq 0}, \tag{7.109b}$$

it is possible to define a linear combination of the measurements (7.107) that is independent of $\chi_0^\circ = \chi^\circ(0)$ as

$$\begin{aligned}
\sum_{h=0}^{\nu} \hat{a}_h \eta_p((k+h)\tau_s) &= \sum_{h=0}^{\nu} \hat{a}_h C_2 \chi^\circ((k+h)\tau_s) + \sum_{h=0}^{\nu} \hat{a}_h \bar{M}_{\eta_p,w} w((k+h)\tau_s) \\
&= C_2 \rho_{A_d}(A_d)\chi^\circ(k\tau_s) + \bar{M}_{\eta_p,w}\rho_{A_d}(S_d) w(k\tau_s) \\
&= \bar{M}_{\eta_p,w}\rho_{A_d}(S_d) w(k\tau_s).
\end{aligned} \tag{7.110}$$

Defining

$$H(k\tau_s) = \begin{bmatrix} \eta_p((k+\nu)\tau_s) & \eta_p((k+\nu-1)\tau_s) & \cdots & \eta_p((k+1)\tau_s) & \eta_p(k\tau_s) \end{bmatrix}, \tag{7.111a}$$

$$\hat{a} = \begin{bmatrix} \hat{a}_\nu & \hat{a}_{\nu-1} & \cdots & \hat{a}_1 & \hat{a}_0 \end{bmatrix}, \tag{7.111b}$$

$$\tilde{M} = \bar{M}_{\eta_p,w}\rho_{A_d}(S_d), \tag{7.111c}$$

yields

$$H(k\tau_s)\hat{a} = \tilde{M}\chi(k\tau_s). \tag{7.112}$$

In turn, (7.112) can be vectorized [49, Sec. 2.5] introducing $\hat{M} = \mathrm{vec}(\tilde{M})$, yielding:

$$H(k\tau_s)\hat{a} = (w(k\tau_s)' \otimes I_{q_p})\hat{M}, \tag{7.113}$$

and collecting (7.113) for $k = 0, 1, \ldots, K-1$ yields (7.104).

Since the degree of the minimal polynomial of a matrix never exceeds the degree of its characteristic polynomial, the above reasoning proves that a vector $v \in \mathbb{R}^{n+1+q_p n_e}$ as required at line 2 of Algorithm 3 can be found.

As for the invertibility of the matrix at line 4, note that this matrix is exactly $\rho_{A_d}(S_d)$. Since S is diagonalizable, there exists an invertible matrix T_I such that $S = T_I \Lambda_S T_I^{-1}$, with Λ_S diagonal. Since τ_s is non-pathological, S_d is diagonalizable by using the same T_I as S, yielding $S_d = T_I \Lambda_{S_d} T_I^{-1}$ with Λ_{S_d} diagonal. By rewriting

$$\rho_{A_d}(S_d) = \rho_{A_d}(T_I \Lambda_{S_d} T_I^{-1}) = T_I \rho_{A_d}(\Lambda_{S_d}) T_I^{-1}, \tag{7.114}$$

and considering that $\rho_{A_d}(\Lambda_{S_d})$ is diagonal, it is clear that $\rho_{A_d}(S_d)$ is not invertible if and only if one of the elements on its diagonal is zero; but this would imply that S_d and A_d have at least one eigenvalue in common, which is impossible since by hypothesis τ_s is non-pathological and $\mathrm{spec}(S) \cap \mathrm{spec}(A) = \emptyset$ due to Assumption 7.8.

To prove item (ii), note that $\tilde{\eta}_p$ is a q_p−dimensional vector function whose scalar components are linear combinations of the n_e linearly independent functions generated by the exosystem; hence, for any fixed $w_0 \in \mathscr{W}^\circ$, the ensuing $\tilde{\eta}_p$ is uniquely associated to a set of $q_p n_e$ scalars. Consequently, if Algorithm 3 is able to determine a matrix $\bar{M}$ (possibly different from the *true* $\bar{M}_{\eta_p,w}$ solving (7.76b)) which ensures that $\bar{M}w$ interpolates $\tilde{\eta}_p$ at $q_p n_e$ (or more) points in time, then for the fixed w_0 the same $\bar{M}$ interpolates $\tilde{\eta}_p$ at any time, since it matched all the $q_p n_e$ degrees of freedom in q_p. Since Algorithm 3 is based on satisfying (7.112) at $K \geq n+1+q_p n_e > q_p n_e$ sample points, the claim in item (ii) is proved once it is clear that (7.112) expresses a matching condition for the relation $\tilde{\eta}_p(t) = \bar{M}_{\eta_p,w} w(t)$ at time $t = k\tau_s$. To cancel out the transient terms in (7.107), a linear combination of responses has been performed in (7.110), hence (7.112) describes the steady-state response due to the initial state $\bar{w}_0 = \rho_{A_d}(w_d) w_0$ (see (7.110)). Note that $\bar{w}_0$ excites in S_d all and only the natural modes excited by w_0 in S_d, which implies that the coefficients of such modes are exactly estimated.

To prove item (iii), denote by T_w the block depending on w in (7.103). Note that T_w is full column rank if and only if

$$\operatorname{rank}\left(\begin{bmatrix} w(0) & w(\tau_s) & \cdots & w((K-1)\tau_s) \end{bmatrix}\right) = n_e. \tag{7.115}$$

Since $K > n_e$ and each element $w(\tau_s)$ can be expressed as $w(\tau_s) = S_d^k w_0$, Cayley-Hamilton's theorem implies that (7.115) is equivalent to:

$$\operatorname{rank}\left(\begin{bmatrix} w_0 & S_d w_0 & \cdots & S_d^{n_e-1} w_0 \end{bmatrix}\right) = n_e. \tag{7.116}$$

The condition in (7.116) has the form of a reachability matrix test for the pair (S_d, w_0), which is equivalent to w_0 exciting all natural modes of S_d if S_d has all distinct eigenvalues. By Assumption 7.10, S_d has all distinct eigenvalues since, by Assumption 7.8, S has all distinct eigenvalues; this implies, in turn, that S and S_d have the same eigenvectors, so that w_0 excites all natural modes of S_d if and only if it excites all natural modes of S. Hence, T_w is full column rank by the condition in item (iii).

Let T_{η_p} denote the block depending on η_p in (7.103), and by $T_{\eta_p}^h$ the submatrix containing its rightmost h columns. Since the columns of T_w are a linearly independent set, the algorithm starts to consider matrices $[T^h_{\eta_p}\ \ T_w]$ increasing h until for $h = \nu$, it happens that $[T^{\nu-1}_{\eta_p}\ \ T_w]$ is still full column rank, whereas $[T^{\nu}_{\eta_p}\ \ T_w]$ has a non-trivial kernel, which has necessarily dimension one since $[T^{\nu}_{\eta_p}\ \ T_w]$ has exactly one column more than $[T^{\nu-1}_{\eta_p}\ \ T_w]$. Thus, there is a unique solution υ to (7.104) with the structure in (7.105) and minimal value of ν. Since a solution (7.106) is found in item (i) on the basis of (7.105) and since the latter is shown to be uniquely defined, under the stated hypotheses the unique solution found by Algorithm 3 must result in the same $\bar{M}$ arising from (7.76b). $\square$

Remark 7.15. *Although Algorithm 3 is provided in this context with the specific purpose of estimating $\bar{M}_{\eta_p,w}$ to compute the dynamic allocator* (7.75)*, it is a general-purpose algorithm for the estimation of the pseudo-steady-state output map of a linear system such as,* e.g.*, system* (C.1)-(C.6) *in Appendix C. Thus, its interest falls within the broader context of online data-driven moment estimation for linear systems, as do the statements of Proposition 7.2 and its proof requiring only an adapted notation.* ▲

Algorithm 3 provides a procedure to compute in finite time the information summarized in the item **K1**, for any $\mu \in \mathcal{M}$. Conversely, computing **K2** entails estimating the moments $W_{\eta_p v}(0), W_{\eta_p v}(j\omega_i)$, $\omega_i \in \mathrm{spec}(S)$, $i = 1, \ldots, \bar{n}_e$. In this case, however, the measurements of $\eta_p(t)$ are not influenced only by $v(t)$ but also by $w(t)$ (since the exosystem cannot be disconnected at any given time). Thanks to the computation of **K1**, $\bar{M}_{\eta_p,w}$ is now available. The next statement follows trivially from the previous results.

Lemma 7.5. *If **K1** is available, then Algorithm 4 computes an estimate of **K2**.* $\circ$

Finally, the key step to determine **K3** is to find a polynomial annihilator $N_a(s)$ such that its columns are a minimal polynomial base of the null space of $W_{\eta_o v}(s)$. It is then

Algorithm 4 Data-driven computation of **K2**.

Input: $\bar{M}_{\eta_p,w}$, Ω, Λ, measurements of $\eta_p(t)$.
Output: $W_{\eta_p v}(0), W_{\eta_p v}(j\omega_i)$, $\omega_i \in \operatorname{spec}(S)$, $i = 1, ..., \bar{n}_e$.

1: Define

$$\eta_p^\delta(t) = \eta_p(t) - \bar{M}_{\eta_p,w}\, w(t), \quad t \in \mathbb{R}_{\geq 0}. \tag{7.117}$$

2: Define the *virtual dynamics*

$$\dot{\zeta}(t) = \Omega\zeta(t), \qquad t \in \mathbb{R}_{\geq 0}, \tag{7.118a}$$

$$\psi(t) = \Lambda\zeta(t), \qquad t \in \mathbb{R}_{\geq 0}, \tag{7.118b}$$

with $\zeta(0) = \Lambda'$.

3: **for** $i = 1, \ldots, m$ **do**
4: Apply $v(t) = e_i\psi(t)$, $t \in \mathbb{R}_{\geq 0}$, and measure $\eta_p^\delta(t)$.
5: Apply Algorithm 3 to the measurements of $(\zeta(t), \eta_p^\delta(t))$, yielding

$$\bar{M}_i = \begin{bmatrix} \bar{M}_{i,0} & \bar{M}_{i,1} & \ldots & \bar{M}_{i,2\bar{n}_e} \end{bmatrix}, \tag{7.119}$$

with $\bar{M}_{i,h} \in \mathbb{R}^{q_p}$, $h = 0, \ldots, 2\bar{n}_e$.

6: The i-th column of $W_{\eta_p v}(0)$ is $\bar{M}_{i,0}$.
7: The i-th column of $W_{\eta_p v}(j\omega_h)$ is $\bar{M}_{i,2h-1} + j\bar{M}_{i,h}$, $h = 1, \ldots, \bar{n}_e$.
8: **end for**

transformed into a rational annihilator $W_a(s)$ by adding a Hurwitz denominator to make it proper. Since the numerators in the elements of $W_{\eta_o v}(s)$ have degree at most n, then, according to [50, Prop. 1], $N_a(s)$ can be chosen with elements of degree not larger than nq_o exactly by selecting the columns of $N_a(s)$ as a minimal polynomial base of the null space of $W_{\eta_o v}(s)$ [51, Main Thm.].

Define

$$n_c = nq_o + 1, \tag{7.120a}$$

$$n_r = \min\{v \in \mathbb{N} : 2vq_o \geq mn_c\}. \tag{7.120b}$$

For a set of frequencies $\{\bar{\omega}_1, ..., \bar{\omega}_{n_r}\}$, define the block matrix H with the (k, h)-th block of size $2q_o \times m$ as

$$H_{[k,h]} = \begin{bmatrix} \mathrm{Re}\left((j\bar{\omega}_k)^{h-1} W_{\eta_o v}(j\bar{\omega}_k)\right) \\ \mathrm{Im}\left((j\bar{\omega}_k)^{h-1} W_{\eta_o v}(j\bar{\omega}_k)\right) \end{bmatrix}, \quad k = 1, \dots, n_r,\ h = 1, \dots, n_c, \tag{7.121}$$

where $\mathrm{rank}(W_{\eta_o v}(j\bar{\omega}_k)) = q_o$ provided that $s = j\bar{\omega}_k$ is not a pole or a zero of the transfer matrix $W_{\eta_o v}(s)$.

The computation of a polynomial annihilator for $W_{\eta_o v}(s)$ can be translated into the computation of the null space of the real matrix H. The key issue is that, while every polynomial vector in a minimal basis of the null space of $W_{\eta_o v}(s)$ is associated to a single real vector in the kernel of H, the opposite transformation maps a basis of $\ker(H)$ to a linearly dependent set of polynomial vectors in the null space of $W_{\eta_o v}(s)$. This problem is avoided by computing a *minimal coefficient basis* for the null space of H, whose columns, when translated back into their polynomial form, yield a *minimal polynomial basis* for the null space of $W_{\eta_o v}(s)$ with dimension $\rho = m - q_o$. The following algorithm is adapted from [50], where several possible optimized computations are also discussed. A minimal *polynomial* basis matrix of order $h-1$ is computed from a minimal *coefficient* basis matrix $N \in \mathbb{R}^{mh \times \rho}$ as

$$\begin{bmatrix} I_m & sI_m & \cdots & s^{h-1} I_m \end{bmatrix} N. \tag{7.122}$$

Proposition 7.3. *Recall* (7.120)*. Consider the transfer matrix* $W_{\eta_o v}(s) \in \mathbb{C}^{q_o \times m}$ *and an arbitrary collection of frequencies* $\{\bar{\omega}_1, \dots, \bar{\omega}_n\}$ *such that* $\mathrm{rank}(W_{\eta_o v}(j\bar{\omega}_i)) = q_o$*,* $i = 1, \dots, n$*. Let the matrix* $H \in \mathbb{R}^{q_o n_r \times m n_c}$ *be defined as in* (7.121)*. Let* $N \in \mathbb{R}^{mh \times (m - q_o)}$ *be a* minimal coefficient basis *matrix computed by Algorithm 5 and partitioned as*

$$N = \begin{bmatrix} N_0 \\ N_1 \\ \vdots \\ N_{h-1} \end{bmatrix}, \quad N_i \in \mathbb{R}^{m\times(m-q_o)},\ i = 0,\ldots,h-1. \tag{7.123}$$

Then, the matrix function

$$N_a(s) = N_0 + sN_1 + \ldots + s^{h-1}N_{h-1} \tag{7.124}$$

is a polynomial annihilator of $W_{\eta_o v}(s)$ whose columns are a minimal polynomial basis for the null space of $W_{\eta_o v}(s)$. ∘

Proof. Since each element in the polynomial matrix $\det(sI-A)W_{\eta_o v}(s)$ has degree not larger than n and $\mathrm{rank}(\det(sI-A)W_{\eta_o v}(s)) = \eta_o$ by Assumption 7.8, it follows from [50, Prop. 1] that a polynomial annihilator $N_a(s)$ with entries having maximum degree not larger than nq_o exists. Then, the equation $W_{\eta_o v}(s)N_a(s) = 0$ has a solution of the form

$$N_a(s) = N_0 + sN_1 + \ldots + s^{\bar{n}}N_{\bar{n}}, \quad \bar{n} < n_c. \tag{7.125}$$

Let

$$N_{\bar{n}} = \begin{bmatrix} N_0 \\ N_1 \\ \vdots \\ N_{\bar{n}} \end{bmatrix} \tag{7.126}$$

so that

$$N_a(s) = \begin{bmatrix} I_m & sI_m & \cdots & s^{\bar{n}}I_m \end{bmatrix} N_{\bar{n}}. \tag{7.127}$$

Thus, the annihilator equation to be solved becomes

$$W_{\eta_o v}(s) N_a(s) = W_{\eta_o v}(s) \begin{bmatrix} I_m & sI_m & \dots & s^{\bar{n}} I_m \end{bmatrix} N_{\bar{n}} = 0. \qquad (7.128)$$

By evaluating (7.128) at the frequencies $\{\bar{\omega}_1, \dots, \bar{\omega}_n\}$ and imposing that both real and imaginary parts of the left hand side of (7.128) are equal to zero, it follows that $N_{\bar{n}}$ lies in the null space of the matrix $H_{[:,1:\bar{n}]}$ whose blocks are defined in (7.121). Finally, bu arguments similar to those in [50], Algorithm 5 computes the minimal $\bar{n} = h$ and a suitable $N_{\bar{n}} = N$, yielding the desired result. □

Remark 7.16. *The requirement of selecting frequencies* $\{\bar{\omega}_i\}_{i=1}^{n}$ *with a full-rank property, namely such that* $\mathrm{rank}(W_{\eta_o v}(j\bar{\omega}_i)) = q_o$*,* $i = 1, \dots, n$*, originates from the necessity to rule out pathological cases that may arise in control systems such as those modeled by* (7.59)*. More precisely, suppose that* (7.59) *embeds a feedback control loop (recall the linear case of Section 7.6.1), which has been designed by incorporating an internal model of the exosystem, as is customary in output regulation problems. In that case, choosing frequencies that belong to the internal model yields null rows in* H*, making the polynomial annihilator's computation unfeasible. It is worth observing that, in the uncertain setting, a random selection of such frequencies* generally *ensures the required full-rank property.* ▲

Remark 7.17. *Algorithm 5 can be applied to find a minimal coefficient basis for a polynomial annihilator of the polynomial matrix* $A(s) = s^{\bar{n}} A_{\bar{n}} + \dots + sA_1 + A_0$ *where* $A_i \in \mathbb{R}^{q_o \times m}$ *and* $\mathrm{rank}(A(s)) = q_o$*. In such a case, instead of forming matrix* H *with blocks described by* (7.121)*, it is possible to use a block Toeplitz matrix* T *whose* $(k,h)-$*th block is* $T_{[k,h]} = A_{k-h}$ *if* $0 \le k - h \le \bar{n}$*, and* $T_{[k,h]} = 0$ *otherwise. This applies to,* e.g.*, the case of* $\mu = 0$*, in which the polynomial matrix* $N_{\eta_o v}(s) = \det(sI - A) W_{\eta_o v}(s)$ *is known.* ▲

Recalling the definition of matrix H in (7.121), it appears evident that Proposition 7.3

is stated with the ideal premise of a perfect knowledge of the transfer matrix $W_{\eta_o v}(s)$ evaluated at the frequencies $\{\bar{\omega}_1, \ldots, \bar{\omega}_n\}$; such a knowledge is not available for a generic $\mu \neq 0$. Therefore, to extend the previous results to the case of a generic $\mu \in \mathcal{M}$, it is necessary to perform a data-driven estimation of the moments $W_{\eta_o v}(j\bar{\omega}_i)$, $i = 1, \ldots, n$. The proposed procedure is illustrated in Algorithm 6.

The following result summarizes all the previous claims, and proves that the proposed control allocator is determined for any $\mu \in \mathcal{M}$.

Theorem 7.3. *Consider the system* (7.59)-(7.60) *with* $\mu \in \mathcal{M}$. *Suppose that Assumptions 7.4, 7.8, 7.9 and 7.10 hold. Then, the dynamic control allocator* (7.75) *interconnected to* (7.59)-(7.60) *as in* (7.63) *solves Problem 7.5 for any* $\gamma \in \mathbb{R}_{>0}$. ∘

Proof. The validity of Theorem 7.3 follows from the combination of all the previous results. Indeed, the full design procedure for any $\mu \in \mathcal{M}$ can now be illustrated, providing a constructive proof. Consider the data **K1**, **K2**, **K3** in Definition 7.1. If $\mu = 0$, *i.e.*, (7.59) is fully known, then Remark 7.10 provides a way to compute all the items. Conversely, if $\mu \neq 0$, it is necessary to perform an *online estimation* of all the data items. The procedure is summarized as follows.

1. **K1** is determined using Algorithm 3, whose validity is proven by Proposition 7.2.

2. **K2** is given by Algorithm 4, whose correctness is guaranteed by Lemma 7.5.

3. **K3**, *i.e.*, the polynomial annihilator of (7.59) with respect to the regulated output, is determined by first computing the matrix H in (7.121) using Algorithm 6, then using it in Algorithm 5 to obtain a minimal polynomial basis for the null space of $W_{\eta_o v}(s)$, as stated in Proposition 7.3.

Once the data **K1**, **K2**, and **K3** have been estimated, it is possible to build the control allocator (7.75), which solves Problem 7.5 for any $\gamma \in \mathbb{R}_{>0}$ as stated in Lemma 7.4. □

The results hereby described are now corroborated by a numerical example. The dynamics (7.59) are now interpreted as the output feedback interconnection of a plant and a controller, whose state vectors are denoted by $x(t) \in \mathbb{R}^{n_p}$ and $\varphi(t)\mathbb{R}^{n_c}$, respectively. This situation is similar to the one addressed by the linear case of Section 7.6.1. Thus, the matrices in (7.59) are decomposed into

$$A(\mu) = \begin{bmatrix} A_p(\mu) & B_p(\mu)C_c \\ B_c C_p(\mu) & A_c \end{bmatrix}, \quad B(\mu) = \begin{bmatrix} B_p(\mu) \\ 0 \end{bmatrix}, \quad P(\mu) = \begin{bmatrix} 0 \\ B_c Q \end{bmatrix}. \tag{7.131}$$

The nominal values for the plant are

$$A_p(0) = \begin{bmatrix} 1 & -1 & 2 \\ 0 & 1 & -1 \\ -1 & 1 & 1 \end{bmatrix}, \quad B_p(0) = \begin{bmatrix} 1 & 1 \\ -1 & 0 \\ 2 & 1 \end{bmatrix}, \quad C_p(0) = \begin{bmatrix} 1 & 1 & 0 \end{bmatrix}, \tag{7.132}$$

and $Q = [-1 \;\; 0]$. The exosystem is defined by (7.60) with $n_e = 2$ and $\omega_1 = 1$.

By relying on a classic architecture, the design of the controller comprises an internal model unit Θ_1, which contains a single copy (namely, as the number of regulated outputs) of the modes of the exosystem, and a dynamic stabilizer Θ_2, which contains an observer for the cascade of the plant and Θ_1 and a stabilizing feedback from the observer dynamics. In particular, a state-space realization of Θ_1 is provided by the matrices $A_{\Theta_1} = S$, $C_{\Theta_1} = [1 \;\; 0]$ and $B_{\Theta_1} = I_2$, while the interconnection between Θ_1 and the system (7.132) is achieved by considering a constant *squaring-down* matrix

$\Xi = [1 \;\; 2]'$. The state-space description of the compensator Θ_2 is

$$\dot{x}_{\Theta_2}(t) = (A_{\Theta_2} + B_{\Theta_2}K_{\Theta_2} - L_{\Theta_2}C_{\Theta_2})x_{\Theta_2}(t) + L_{\Theta_2}\eta_o(t), \quad t \in \mathbb{R}_{\geq 0}, \tag{7.133}$$

together with the interconnection given by

$$u_{\Theta_1}(t) = y_{\Theta_2}(t) = K_{\Theta_2}x_{\Theta_2}(t), \quad t \in \mathbb{R}_{\geq 0}, \tag{7.134}$$

where

$$A_{\Theta_2} = \begin{bmatrix} A_p(0) & B_p(0)\Xi C_{\Theta_1} \\ 0 & A_{\Theta_1} \end{bmatrix}, \quad B_{\Theta_2} = \begin{bmatrix} 0 \\ I \end{bmatrix}, \quad C_{\Theta_2} = \begin{bmatrix} C_p(0) & 0 \end{bmatrix}. \tag{7.135}$$

The gain matrices

$$L_{\Theta_2} = \begin{bmatrix} 10.85 & -3.51 & 12.25 & 1.40 & -0.17 \end{bmatrix}', \tag{7.136a}$$

$$K_{\Theta_2} = \begin{bmatrix} -8.09 & -4.41 & -3.40 & -8.07 & -0.92 \\ -0.41 & -0.89 & -0.44 & -0.92 & -1.41 \end{bmatrix}, \tag{7.136b}$$

have been obtained with an LQR approach. Finally, the regulated and performance outputs are defined as in (7.59b) and (7.59c), respectively, with

$$C_1(\mu) = \begin{bmatrix} C_p(\mu) & 0 \end{bmatrix}, \quad C_2(\mu) = \begin{bmatrix} 0 & C_c \end{bmatrix}, \quad C_c = \begin{bmatrix} C_{\Theta_1} & 0 \end{bmatrix}, \tag{7.137}$$

and $D_{21} = 0$, $D_{12} = Q$, $D_{21} = I$, $D_{22} = 0$.

A polynomial annihilator of the nominal transfer matrix $W_{\eta_o v}(s)$ is

$$N_a(s) = \begin{bmatrix} -s^2 - s - 3 \\ -3s - 3 \end{bmatrix}, \quad s \in \mathbb{C}, \tag{7.138}$$

hence, its state-space description is given by

$$A_a = \begin{bmatrix} -3 & -3 & -1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}, \quad B_a = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \quad C_a = \begin{bmatrix} -1 & -1 & -3 \\ 0 & -3 & -3 \end{bmatrix}. \tag{7.139}$$

The optimization unit is designed as in (7.75) with $R = I$ for multiple values of γ.

Algorithm 5 Computation of a minimal coefficients basis.

Input: $H = \begin{bmatrix} H_{[:,1]} & H_{[:,2]} & \dots \end{bmatrix}$.
Output: A minimal coefficient basis N for H.
1: $\bar{\Sigma} \leftarrow H_{[:,1]}$
2: $k_1 \leftarrow \dim(\ker(\bar{\Sigma}))$
3: $\bar{p} \leftarrow \emptyset$
4: $h \leftarrow 1$
5: **if** $k_1 = 0$ **then**
6: $\quad X_2 \leftarrow 0_{1\times(2m)}$
7: $\quad h \leftarrow 2$
8: $\quad$ **goto** 15
9: **else if** $0 < k_1 < \rho$ **then**
10: $\quad X_1 \leftarrow 0_{1\times m}$
11: $\quad$ **goto** 22
12: **else if** $k_1 = \rho$ **then**
13: $\quad$ **goto** 29
14: **end if**
15: $\bar{\Sigma} \leftarrow \begin{bmatrix} H'_{[:,1:h]} \\ X'_h \end{bmatrix}$
16: $k_h \leftarrow \dim(\ker(\bar{\Sigma}))$
17: **if** $k_h = \rho$ **then**
18: $\quad$ **goto** 29
19: **else**
20: $\quad$ **goto** 22
21: **end if**
22: $\bar{Q}, \bar{R} \leftarrow$ QR decomposition of $\bar{\Sigma}'$.
$\triangleright$ $\bar{\Sigma}' = \bar{Q}\bar{R}$ with $\bar{Q}$ orthogonal and $\bar{R}$ upper-triangular.
23: $\bar{L} \leftarrow$ The last $(p - \text{rank}(\bar{R}))$ rows of $\bar{Q}'$.
24: $\bar{p} \leftarrow \bar{p} \cup \bar{p}_h$, where $\bar{p}_h$ is the set of pivot indices in a reduced row echelon form of $\bar{L}$.
25: $\bar{X} \leftarrow$ The rows of I_m with indices in $\bar{p}$.
26: $X_{h+1} \leftarrow \text{diag}(X_h, \bar{X})$
27: $h \leftarrow h + 1$
28: **goto** 15
29: $N \in \mathbb{R}^{mh\times\rho} \leftarrow$ Any basis matrix of $\ker(\bar{\Sigma})$.

Algorithm 6 Computation of H in (7.121).

Input: $\{\bar{\omega}_i\}_{i=1}^n$, τ_s, $K \in \{a \in \mathbb{N} : a > K > n + n_e + 1\}$.
Output: H.

Define the *virtual dynamics*

$$\bar{\Omega} = \operatorname{diag}\left(\begin{bmatrix} 0 & \bar{\omega}_1 \\ -\bar{\omega}_1 & 0 \end{bmatrix}, \ldots, \begin{bmatrix} 0 & \bar{\omega}_n \\ -\bar{\omega}_n & 0 \end{bmatrix}\right), \tag{7.129a}$$

$$\bar{\Lambda} = \left[\begin{array}{c|c|c} 1\ 0 & \cdots & 1\ 0 \end{array}\right], \tag{7.129b}$$

$$\dot{\zeta}(t) = \bar{\Omega}\zeta(t), \qquad t \in \mathbb{R}_{\geq 0}, \tag{7.129c}$$

$$\psi(t) = \bar{\Lambda}\zeta(t), \qquad t \in \mathbb{R}_{\geq 0}, \tag{7.129d}$$

with $\zeta(0) = \bar{\Lambda}'$.

$v(t) \leftarrow 0$, $t \in \mathbb{R}_{\geq 0}$

Collect samples $\mathscr{W}_d = \{w(k\tau_s)\}_{k=0}^{n+K-1}$, $\mathscr{H}_d = \{\eta_o(k\tau_s)\}_{k=0}^{n+K-1}$.

Apply Algorithm 3 to $(\mathscr{W}_d, \mathscr{H}_d)$ to obtain $\bar{M}_{\eta_o,w}$.

$\eta_o^\delta(t) \leftarrow \eta_o(t) - \bar{M}_{\eta_o,w} w(t)$

for $i = 1, \ldots, m$ **do**

$v(t) \leftarrow e_i \psi(t)$, $t \in \mathbb{R}_{\geq 0}$

$\triangleright$ Connect (7.129) to the i-th input channel.

Collect samples $\mathcal{Z}_d = \{\zeta(k\tau_s)\}_{k=0}^{n+K-1}$, $\bar{\mathscr{H}}_d = \{\eta_o^\delta(k\tau_s)\}_{k=0}^{n+K-1}$.

Apply Algorithm 3 to $(\mathcal{Z}_d, \bar{\mathscr{H}}_d)$ and obtain $\bar{M}^i_{\eta_o,\zeta}$ partitioned as

$$\bar{M}^i_{\eta_o,\zeta} = \begin{bmatrix} \operatorname{Re}\left(W^i_{\eta_o v}(j\bar{\omega}_1)\right)' \\ \operatorname{Im}\left(W^i_{\eta_o v}(j\bar{\omega}_1)\right)' \\ \vdots \\ \operatorname{Re}\left(W^i_{\eta_o v}(j\bar{\omega}_n)\right)' \\ \operatorname{Im}\left(W^i_{\eta_o v}(j\bar{\omega}_n)\right)' \end{bmatrix}'. \tag{7.130}$$

end for

Rearrange the columns of $\{\bar{M}^i_{\eta_o,\zeta}\}_{i=1}^m$ to obtain H as in (7.121).

C Moments of a linear system

The contents of this Appendix are based on [42, 43] and references therein. The interested reader is referenced to those works for a detailed discussion of the following topics. Consider the system

$$\dot{x}(t) = Ax(t) + Bu(t), \qquad t \in \mathbb{R}_{\geq 0}, \tag{C.1a}$$

$$y(t) = Cx(t) + Du(t), \qquad t \in \mathbb{R}_{\geq 0}, \tag{C.1b}$$

where $x(t) \in \mathbb{R}^n$ is the state vector, $u(t) \in \mathbb{R}^m$ is the input vector, and $y(t) \in \mathbb{R}^q$ is the output vector. $A \in \mathbb{R}^{n\times n}$, $B \in \mathbb{R}^{n\times m}$, $C \in \mathbb{R}^{q\times n}$, $D \in \mathbb{R}^{q\times m}$ are constant matrices. The transfer matrix of (C.1) is

$$W(s) = C(sI - A)^{-1}B + D, \quad s \in \mathbb{C}. \tag{C.2}$$

Definition C.1 (Moment)**.** *Let* $s^\star \in \mathbb{C}$ *be such that* $s^\star \notin \mathrm{spec}(A)$ *for the system* (C.1). *The* moment of (C.1) at $s^\star$ *is* $W(s^\star)$. ∘

The *moment of* $W(s)$ *(or* (A, B, C, D)*) at* $s^\star$ has the following dynamic meaning: under the assumption that $s^\star \notin \mathrm{spec}(A)$, if (C.1) is subject to the input $U_0 e^{s^\star t}$ with $U_0 \in \mathbb{C}^m$, then its output will contain, among others, one term of the form $Y_0 e^{s^\star t}$ with $Y_0 = W(s^\star)U_0$.

Moments can be computed by using only real matrix computations, via the solution of the linear equations

$$\Pi S = A\Pi + BL, \tag{C.3a}$$

$$M = C\Pi + DL. \tag{C.3b}$$

in the unknowns Π, M, provided that

$$S = \begin{bmatrix} \alpha I_m & \omega I_m \\ -\omega I_m & \alpha I_m \end{bmatrix}, \qquad L = \begin{bmatrix} I_m & 0_{m\times m} \end{bmatrix}, \qquad \text{if } s^\star \notin \mathbb{R}, \tag{C.4a}$$

$$S = \alpha I_m, \qquad L = I_m, \qquad \text{if } s^\star \in \mathbb{R}, \tag{C.4b}$$

where $\alpha = \mathrm{Re}(s^\star)$, $\omega = \mathrm{Im}(s^\star)$. Under such conditions, it holds that M satisfies

$$M = \begin{bmatrix} \mathrm{Re}(W(s^\star)) & \mathrm{Im}(W(s^\star)) \end{bmatrix}, \qquad \text{if } s^\star \notin \mathbb{R}, \tag{C.5a}$$

$$M = W(s^\star), \qquad \text{if } s^\star \in \mathbb{R}, \tag{C.5b}$$

i.e., knowing M is equivalent to knowing $W(s^\star)$. Since $s^\star \notin \mathrm{spec}(A)$ implies $\mathrm{spec}(S) \cap \mathrm{spec}(A) = \emptyset$, by standard results [49, Lemma 2.7] the Sylvester equation (C.3a) is solvable and has a unique solution.

Remark C.1. *While the relation between* (C.3) *and* (C.5) *was discussed focusing for simplicity on the computation of one moment of $W(s)$ at a single $s^\star \in \mathbb{C}$, similar relations hold as well for the simultaneous computation of the moments at several complex values, choosing $S \in \mathbb{R}^{n_e \times n_e}$ as a block diagonal matrix $S = \mathrm{diag}(S_1, S_2, \ldots)$ and $L \in \mathbb{R}^{m\times n_e}$ as a block row matrix $L = \mathrm{row}(L_1, L_2, \ldots)$, each having one block for each different $s_1^\star$, $s_2^\star, \ldots$, and each pair of blocks (S_i, L_i) having the structure of the pair (S, L) in* (C.4). ▲

More in general, considering the dynamics

$$\dot{w}(t) = Sw(t), \qquad t \in \mathbb{R}_{\geq 0}, \tag{C.6a}$$

$$\dot{x}(t) = Ax(t) + Pw(t), \qquad t \in \mathbb{R}_{\geq 0}, \tag{C.6b}$$

$$y(t) = Cx(t) + Qw(t), \qquad t \in \mathbb{R}_{\geq 0}, \tag{C.6c}$$

with $S \in \mathbb{R}^{n_e \times n_e}$ and under the assumption that $\mathrm{spec}(S) \cap \mathrm{spec}(A) = \emptyset$, the matrices Π, M solving

$$\Pi S = A\Pi + P, \tag{C.7a}$$

$$M = C\Pi + Q, \tag{C.7b}$$

have the following dynamic interpretation. If the initial condition for (C.6) satisfies $x_0 = \Pi w_0$, then

$$x(t) = \Pi w(t), \quad y(t) = Mw(t), \quad \forall t \geq 0, \tag{C.8}$$

motivating the names of *pseudo-steady-state maps* (respectively, in the state and in the output) for Π and M.

Under the stronger assumption that

$$\mathrm{spec}(A) \subset \mathbb{C}_{<0}, \;\; \mathrm{spec}(S) \subset \mathbb{C}_{\geq 0}, \tag{C.9}$$

which ensure the existence of a steady-state response for (C.6b), (C.6c) under the input generated by (C.6a), matrices Π and M will be called *steady-state maps* (respectively, in the state and in the output). Under (C.9), the steady-state response of (C.6) satisfies the relations

$$\tilde{x}(t) = \Pi w(t), \qquad t \in \mathbb{R}_{\geq 0}, \tag{C.10a}$$

$$\tilde{y}(t) = M w(t), \qquad t \in \mathbb{R}_{\geq 0}, \tag{C.10b}$$

and, for generic initial states $x_0 \neq \Pi w_0$, of (C.6), the corresponding transient response is given by

$$\hat{x}_0 = x_0 - \Pi w_0, \tag{C.11a}$$

$$x^{\circ}(t) = e^{At}\hat{x}_0, \qquad t \in \mathbb{R}_{\geq 0}, \tag{C.11b}$$

$$y^{\circ}(t) = Ce^{At}\hat{x}_0, \qquad t \in \mathbb{R}_{\geq 0}. \tag{C.11c}$$

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